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Forecasting epidemic trajectories: Time Series Growth Curves package tsgc

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Abstract

This paper documents the Time Series Growth Curves (**tsgc**) package for R, which is designed for forecasting epidemics, including the detection of new waves and turning points. The package implements time series growth curve methods founded on a dynamic Gompertz model, which can be estimated using techniques based on state space models and the Kalman filter. The model is suitable for predicting future values of any variable that, when cumulated, is subject to some unknown saturation level. In the context of epidemics, the model can adjust to changes in social behaviour and policy. It is also relevant for many other domains, such as the diffusion of new products. The **tsgc** package is demonstrated using COVID-19 cases and hospitalisations, leading-indicator examples, and lower-frequency growth data.

Keywords: COVID-19, Gompertz growth curve, Kalman filter, reproduction number, state space model, stochastic trend, turning points,.

1. Forecasting epidemic trajectories

Outbreaks of infectious diseases with epidemic potential require real-time responses by public health authorities. Accurate real-time forecasting of the trajectory of the epidemic over the near future is of great value in this regard.

The R package **tsgc** is intended for monitoring and forecasting the progress of an epidemic, including detecting *new waves* and *turning points*.[1] It develops and implements time series growth curve methods first reported in Harvey and Kattuman (2020) (hereinafter referred to as HK). HK develop a class of time series models for predicting future values of a variable which,

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[1]The released package is available on CRAN; the development version is available from GitHub. See Section 7 for installation and worked examples.

when cumulated, is subject to an unknown saturation level. In a single wave of an epidemic, as more and more people get infected, the pool of susceptible individuals dwindles. This results in the decline of new infections, and the cumulative number of infections approaches its saturation level. The model can take account of deviations relative to this canonical trajectory due to changes in social behaviour and policy. Models in this family are relevant for many other disciplines, such as marketing (when estimating the demand for new products). While attention here is focused on the spread of epidemics and the applications used for illustration relate to coronavirus, this package is designed with a view to wider applicability.

Given the number of different modelling approaches for epidemics, software for epidemic monitoring is best understood as an ecology of methods that differ in the amount of epidemiological structure they impose, and the inferential task they support. At one end are mechanistic compartmental, meta-population, network, and agent-based models, which seek to represent explicitly the processes by which infection spreads through a population. Packages such as **EpiModel** support deterministic compartmental, stochastic individual-contact, and network epidemic models (Jenness, Goodreau, and Morris 2018), while bespoke real-time systems, such as the PHE/Cambridge model, use age- and region-structured SEIR-type formulations to integrate deaths, serology, contact data, prevalence surveys, vaccination, and hospital data for policy nowcasting and projection (Birrell, Blake, van Leeuwen, Gent, and De Angelis 2021; Birrell, Blake, Kandiah, Alexopoulos, van Leeuwen, Pouwels, Ghosh, Starr, Walker, House, Gay, Finnie, Gent, Charlett, and De Angelis 2026). Less structurally demanding, but still epidemiologically grounded, are semi-mechanistic renewal models, in which current incidence is generated from past incidence through a generation-interval distribution and a time-varying reproduction number. Packages such as **EpiEstim** and **EpiNow2** belong to this class: they estimate transmission indicators such as R_t and latent incidence from observed surveillance data without requiring a full compartmental representation of the population (Cori, Ferguson, Fraser, and Cauchemez 2013; Abbott, Hellewell, Thompson, Sherratt, Gibbs, Bosse, Munday, Meakin, Doughty, Chun, Chan, Finger, Campbell, Endo, Pearson, Gimma, Russell, Flasche, Kucharski, Eggo, and Funk 2020).

A related but distinct layer of the software ecosystem concerns the observation process rather than the transmission process itself. Surveillance data are often right-truncated, delayed, revised, and affected by changes in testing or reporting practice. This has motivated nowcasting packages such as **NobBS** and **epinowcast**, which estimate events that have occurred but have not yet been reported, and can provide delay-adjusted inputs to downstream transmission models, dashboards, or short-term forecasts (McGough, Johansson, Lipsitch, and Menzies 2020; Abbott and Monticone 2021). An operational layer is provided by forecast hubs and ensemble systems, where forecasts from multiple modelling groups are standardised, archived, combined, and evaluated. The COVID-19 Forecast Hub, **Zoltar**, and related software such as **scoringutils** illustrate this shift from individual modelling packages towards an infrastructure for probabilistic forecasting, model comparison, and ensemble construction (Reich, Brooks, Fox, Kandula, McGowan, Moore, Osthus, Ray, Tushman, Yamana, Biggerstaff, Johansson, Rosenfeld, and Shaman 2019; Reich, Cornell, Ray, Hyman, Choi, Johansson, Biggerstaff, Brooks, Drotar, Hallowell, Osthus, and Yamana 2021; Cramer, Huang, Wang, Ray, Cornell, Bracher, Brennen, Castro Rivadeneira, Gerding, House, Jayawardena, Kanji, Khandelwal, Le, Mühlemann, Niemi, Stark, Shah, Wattanachit, Zorn, and Reich 2022; Bosse, Gruson, Cori, van Leeuwen, Funk, and Abbott 2022).

In contrast, the empirical approach implemented in **tsgc** falls into the class of models described as ‘phenomenological’. Although it is motivated by the archetypal pattern in the dynamics of disease spread, it does not rely on structural assumptions derived from epidemiological theory. Avoiding assumptions about infectiousness, severity, and contact patterns is useful when these quantities are difficult to estimate precisely in real time. Our approach makes minimal assumptions and merely requires past observations of the epidemic variable of interest, to which we apply time-series methods to provide predictions over short future time horizons. The model can be estimated quickly and straightforwardly, and subjected to standard diagnostic tests. A statistical model of this type is a useful complement to mechanistic models that attempt to describe the epidemic in terms of underlying processes.

This paper builds on Harvey and Kattuman (2020) but extends that framework in several directions. In particular, it develops a practical reinitialisation procedure for handling successive epidemic waves, introduces a leading-indicator extension that allows related series to improve short-horizon forecasts, embeds the methods in the **tsgc** package together with tools for estimation, forecasting, visualisation, and evaluation, and shows that the same framework can be applied to data observed at frequencies other than daily. Section 2 sets out the state space formulation of the dynamic Gompertz growth curve, derives the predictive recursions used for nowcasting and forecasting, and shows how these quantities can be translated into estimates of the instantaneous reproduction number R_t. Section 3 explains how multiple waves can be accommodated by reinitialising the series and presents a practical rule for triggering reinitialisation. Section 4 introduces the leading-indicator model. Section 5 sets out the approach to forecast evaluation. Section 6 describes the functionality of **tsgc**, Section 7 provides worked illustrations using COVID-19 data, Section 8 discusses the extension of the framework to other data frequencies, and Section 9 concludes.

2. Theory

2.1. Gompertz curve

Our model is based on the sigmoidal growth curve pattern that characterizes epidemics. We start by assuming that the cumulative number of cases follows a Gompertz curve, which is a parsimonious model for the canonical sigmoid shape of cumulative case numbers in a one-wave epidemic. Over the course of a wave, the number of new infected cases increases up to a peak before declining to zero as the pool of susceptible individuals declines. Specifically, if the cumulative number of cases at time t, $\mu(t)$, follows a Gompertz curve, we can write

$$\mu(t) = \bar{\mu} \exp\{\gamma_0 e^{\gamma t}\},$$

where $\bar{\mu}$ is the unknown saturation level for the cumulative number of cases, $\gamma_0 < 0$ is a parameter related to $\mu(0)$ and $\gamma < 0$ is the growth rate parameter. Defining $\dot{\mu}(t) \equiv d\mu(t)/dt$ and $g(t) \equiv \dot{\mu}(t)/\mu(t)$, it is straightforward to show that

$$\ln g(t) = \delta + \gamma t, \tag{1}$$

where $\delta = \ln \gamma_0 \gamma$.

The observational model needs to be specified in discrete, rather than continuous, time. This is straightforward. Let Y_t be the observed cumulative number of cases on day t and

$y_t = Y_t - Y_{t-1}$ be the number of daily new cases. We can then define the growth rate of Y_t as $g_t = y_t/Y_{t-1}$ and replace $\ln g(t)$ with $\ln y_t - \ln Y_{t-1}$.

The dynamic Gompertz state-space specification is form-invariant with respect to the sampling frequency. Provided Y_t is a cumulative series observed on a regular time grid, the definitions

$$y_t = Y_t - Y_{t-1}, \qquad g_t = y_t/Y_{t-1},$$

and the associated estimation and forecasting recursions are unchanged. In practice, moving from daily to weekly or monthly observations requires only three adjustments: (i) constructing Y_t and y_t at the chosen sampling frequency; (ii) specifying the seasonal period, where relevant, to match systematic reporting patterns, for example $s = 7$ for day-of-week effects in daily data and typically no seasonal component for weekly data; and (iii) where a reproduction-number interpretation is relevant, expressing the generation interval τ in the same units as the observation interval.

2.2. Dynamic Gompertz model

The deterministic trend implied by (1) is too inflexible for practical time-series modelling of an epidemic. Replacing it with a stochastic trend allows the model to adapt to changes in dynamics during the course of the epidemic. We call this stochastic-trend counterpart of (1) the dynamic Gompertz model. It is a local linear trend model specified as

$$\ln g_t = \delta_t + \varepsilon_t,\ \varepsilon_t \sim NID(0, \sigma_\varepsilon^2),\ t = 2, ..., T, \tag{2}$$

where $\ln g_t = \ln y_t - \ln Y_{t-1}$ and

$$\delta_t = \delta_{t-1} + \gamma_{t-1}, \tag{3}$$

$$\gamma_t = \gamma_{t-1} + \zeta_t,\ \zeta_t \sim NID(0, \sigma_\zeta^2), \tag{4}$$

where the disturbances ε_t and ζ_t are mutually independent, and $NID(0, \sigma^2)$ denotes normally and independently distributed with mean zero and variance σ^2. Note that the larger the signal-to-noise ratio, $q_\zeta = \sigma_\zeta^2/\sigma_\varepsilon^2$, the faster the estimate of the slope parameter, γ_t, which can be interpreted as the growth rate of the growth rate of cumulative cases, changes in response to new observations. Conversely, a lower signal-to-noise ratio induces more smoothness to the estimates. When $\sigma_\zeta^2 = 0$, the trend is deterministic as in (1).

2.3. State space form and estimation

It is convenient to write the dynamic Gompertz model in general state space form:

$$\begin{aligned} \ln g_t &= Z\alpha_t + \varepsilon_t & \qquad \varepsilon_t &\sim NID(0, \sigma_\varepsilon^2) \\ \alpha_{t+1} &= T\alpha_t + R\eta_t & \qquad \eta_t &\sim NID(0, Q) \end{aligned}$$

with

$$\alpha_t = (\delta_t, \gamma_t)',\ Z = (1, 0),\ \eta_t = (0, \zeta_t)',\ T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},\ R = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},\ Q = \begin{pmatrix} 0 & 0 \\ 0 & \sigma_\zeta^2 \end{pmatrix}.$$

This model can be estimated using techniques based on the Kalman filter once a prior is specified. The prior is

$$(\delta_1, \gamma_1)' \sim N(a_1, P_1),$$

where a_1 is a 2×1 vector of prior means and P_1 a 2×2 prior variance matrix. We use a diffuse prior due to the absence of prior information about the epidemic when the model is first estimated: i.e., we set $a_1 = (0,0)'$, $P_1 = \kappa I$, and let $\kappa \to \infty$. Model estimation, including implementation of the diffuse prior, is carried out using the **KFAS** package (Helske 2017).

The Kalman filter outputs estimates of the state vector $(\delta_t, \gamma_t)'$. The estimates at time t conditional on information up to and including time t are denoted $(\hat{\delta}_{t|t}, \hat{\gamma}_{t|t})'$ and given by the contemporaneous filter; the predictive filter estimates the state at time $t+1$ from the same information set, outputting $(\hat{\delta}_{t+1|t}, \hat{\gamma}_{t+1|t})'$.

It may be useful to review past movements of the state vector $(\delta_t, \gamma_t)'$. This can be done using the smoothed estimates $(\hat{\delta}_t, \hat{\gamma}_t)'$, which denotes the estimates of the state vector at time t based on all T observations in the series.

Estimation of the unknown variance parameters (σ_ε^2 and σ_ζ^2) is by maximum likelihood (ML) and is carried out using **KFAS** following the procedure described in Helske (2017). We retain the option of either estimating the signal-to-noise ratio q_ζ, or of fixing it at a plausible value. In practice, for coronavirus applications, we set the value of q_ζ based on experience and judgment, reducing the number of parameters to be estimated by one. Tests for normality and residual serial correlation are based on the standardised innovations, i.e., one-step-ahead prediction errors, $v_t = \ln g_t - \delta_{t|t-1}$, $t = 3, ..., T$.

Daily effects, which are generally quite pronounced in the coronavirus data, can be included in the model as described in the Appendix.

2.4. Forecasts and peak prediction

Forecasts of future observations are obtained from the predictive recursions

$$\begin{aligned} \widehat{g}_{T+\ell|T} &= \exp(\hat{\delta}_{T|T} + \hat{\gamma}_{T|T}\ell), \;\; \ell = 1, 2, .. \\ \widehat{\mu}_{T+\ell|T} &= \widehat{\mu}_{T+\ell-1|T}(1 + \widehat{g}_{T+\ell|T}) \end{aligned}$$

so that

$$\widehat{y}_{T+\ell|T} = \widehat{g}_{T+\ell|T}\widehat{\mu}_{T+\ell-1|T} = Y_T \exp\left(\hat{\delta}_{T+\ell|T}\right) \prod_{j=1}^{\ell-1} \left\{1 + \exp(\hat{\delta}_{T+j|T})\right\} \tag{5}$$

and $\widehat{Y}_{T+\ell|T} = \widehat{\mu}_{T+\ell|T}$; the initial value is $\widehat{\mu}_{T|T} = Y_T$.

For graphical summaries of future incidence, we report state-implied forecast bands obtained by propagating pointwise forecast bands for the latent log growth rate, δ_t, through the deterministic forecasting recursion. Let

$$\delta^L_{T+j|T} = \hat{\delta}_{T+j|T} - c_\alpha \sqrt{P_{\delta,T+j|T}}, \qquad \delta^U_{T+j|T} = \hat{\delta}_{T+j|T} + c_\alpha \sqrt{P_{\delta,T+j|T}},$$

where

$$P_{\delta,T+j|T} = \mathrm{Var}(\delta_{T+j} \mid \mathcal{F}_T)$$

is the j-step-ahead forecast variance of the latent log growth rate, excluding the observation disturbance, and c_α is the relevant standard normal critical value. Since (5) is monotone increasing in each future value of δ_{T+j}, the lower and upper state-implied forecast bands for incidence are computed as

$$\widehat{y}^{L}_{T+\ell|T} = Y_T \exp\left(\delta^{L}_{T+\ell|T}\right) \prod_{j=1}^{\ell-1} \left\{1 + \exp\left(\delta^{L}_{T+j|T}\right)\right\},$$

and

$$\widehat{y}^{U}_{T+\ell|T} = Y_T \exp\left(\delta^{U}_{T+\ell|T}\right) \prod_{j=1}^{\ell-1} \left\{1 + \exp\left(\delta^{U}_{T+j|T}\right)\right\}.$$

These bands should not be interpreted as formal prediction intervals or confidence intervals for the realised future observation $y_{T+\ell}$. They are obtained by transforming marginal uncertainty bands for the latent log growth rate, rather than by integrating over the full joint predictive distribution of future states, future observation disturbances, and the nonlinear cumulative recursion. In particular, under the Gaussian state-space model,

$$\log y_{T+1} \mid \mathcal{F}_T \sim N\left(\log Y_T + \hat{\delta}_{T+1|T}, P_{\delta,T+1|T} + \sigma^2_{\varepsilon}\right),$$

so the one-step predictive distribution of the future observation y_{T+1} is lognormal. The point forecast $\widehat{y}_{T+1|T} = Y_T \exp(\hat{\delta}_{T+1|T})$ is a deterministic plug-in forecast, not a random variable. For horizons $\ell > 1$, the predictive distribution of $y_{T+\ell} \mid \mathcal{F}_T$ is path-dependent because $Y_{T+\ell-1}$ is itself random and depends on earlier future incidence values. We refer to the plotted incidence bands as state-implied forecast bands.

The latent growth rate of new cases, $g_{y,t}$, can be approximated from the continuous-time incidence curve $\mu'(t) = g(t)\mu(t)$, where $\mu(t)$ is the cumulative growth curve and $g(t)$ is its growth rate. Taking logarithms and differentiating gives the filtered estimate

$$\hat{g}_{y,t|t} = \hat{g}_{t|t} + \hat{\gamma}_{t|t}, \tag{6}$$

where $\hat{g}_{t|t} = \exp \hat{\delta}_{t|t}$. The sampling variability of $\hat{g}_{y,t|t}$ is dominated by that of $\hat{\gamma}_{t|t}$ (see Harvey and Kattuman 2021). Therefore when constructing confidence intervals for $\hat{g}_{t|t}$ we treat $\hat{g}_{y,t}$ as if it has a normal distribution centered on $\hat{g}_{y,t|t}$ with variance $\mathrm{Var}(\hat{\gamma}_{t|t})$.

Even when the nowcast $\hat{g}_{y,T|T}$ is positive and daily cases are growing, there will be a saturation level for the cumulative total, Y_t, so long as $\hat{\gamma}_{T|T}$ is negative. The nowcasts of y_t peak when $\hat{g}_{y,t|t} = 0$, which requires $\hat{\gamma}_{t|t}$ to be sufficiently negative to outweigh $\hat{g}_{t|t}$, which is, of course, always positive. This can be seen from the expression for the growth rate of daily cases:

$$\hat{g}_{y,T|T} = \exp \hat{\delta}_{T|T} + \hat{\gamma}_{T|T} = \hat{g}_{T|T} + \hat{\gamma}_{T|T}. \tag{7}$$

When $\hat{\gamma}_{T|T}$ is negative, there is a flattening of the curve and a signaling of an upcoming peak in the trend of y_t. As shown in [HK, p10], the peak in the trend is predicted to be ℓ_T days ahead where[2]

$$\ell_T = \frac{\ln(-\hat{\gamma}_{T|T}) - \hat{\delta}_{T|T}}{\hat{\gamma}_{T|T}} = \frac{\ln(-\hat{\gamma}_{T|T}/\hat{g}_{T|T})}{\hat{\gamma}_{T|T}}, \quad -\hat{g}_{T|T} < \hat{\gamma}_{T|T} < 0.$$

[2]Note the change in sign of γ as compared with HK.

The generation of forecasts is illustrated in Section 7.1.

2.5. Reproduction Number R_t

The path of the epidemic is best tracked by nowcasts and forecasts of $g_{y,t}$, the growth rate of y_t, which are constructed by HK from the filtered estimates in the state space model, (2), (3) and (4). Wallinga and Lipsitch (2007) describe how the estimates of $g_{y,t}$ can be translated into estimates of the instantaneous reproduction number R_t. Harvey and Kattuman (2021) propose

$$\widetilde{R}_{t,\tau} = 1 + \tau g_{y,t|t} \quad \text{or} \quad \widetilde{R}^e_{\tau,t} = \exp(\tau g_{y,t|t}), \tag{8}$$

where τ is the generation interval – the typical number of days between an infected person becoming infected and them transmitting the disease to someone else. Epidemiologists employ a range of statistical techniques to estimate, refine, and update the generation interval estimates (Park, Sun, Abbott, Sender, Bar-on, Weitz, Funk, Grenfell, Backer, Wallinga, Viboud, and Dushoff 2023). In the illustration of **tsgc** in Section 7, we set the generation interval to 4 days. When working with weekly or monthly data, τ must be specified in the same units as the observation interval (e.g., τ=4 days $\equiv$ 4/7 weeks for weekly data).

We report approximate pointwise uncertainty bands for $\widetilde{R}_{t,\tau}$ and $\widetilde{R}^e_{\tau,t}$ by transforming the corresponding bands for $g_{y,t}$. Let

$$\widehat{g}_{y,t|t} = \exp(\hat{\delta}_{t|t}) + \hat{\gamma}_{t|t},$$

and denote the lower and upper pointwise bands for $g_{y,t}$ by $g^L_{y,t}$ and $g^U_{y,t}$. Since the transformations in (8) are monotone in $g_{y,t}$, the corresponding bands are

$$\left[1 + \tau g^L_{y,t},\ 1 + \tau g^U_{y,t}\right]$$

for $\widetilde{R}_{t,\tau} = 1 + \tau g_{y,t}$, and

$$\left[\exp(\tau g^L_{y,t}),\ \exp(\tau g^U_{y,t})\right]$$

for $\widetilde{R}^e_{\tau,t} = \exp(\tau g_{y,t})$. These are approximate pointwise uncertainty bands for the latent reproduction-number summaries implied by the dynamic Gompertz trend, conditional on the specified generation interval τ. They are not Bayesian credible intervals and do not incorporate uncertainty about τ. See Harvey, Kattuman, and Thamotheram (2021) for an application.

The estimates of R_t can be used for tracking and forecasting the epidemic. The nowcasts of y_t peak when $\hat{g}_{y,t|t} = 0$, corresponding to $\widetilde{R}_{t,\tau} = \widetilde{R}^e_{\tau,t} = 1$. Based on (7), predictions of $g_{y,t}$ are given by

$$\hat{g}_{y,T+\ell|T} = \exp \hat{\delta}_{T+\ell|T} + \hat{\gamma}_{T+\ell|T} = \exp(\hat{\delta}_{T|T} + \hat{\gamma}_{T|T}\ell) + \hat{\gamma}_{T|T}, \qquad \ell = 1, 2, . \tag{9}$$

We can then obtain predictions of R_t, as in (8). If $\hat{\gamma}_{T|T}$ is zero, the estimated growth of y_t is exponential and it is helpful to characterize it by the doubling time, $\ln 2/\hat{g}_{y,T|T} = 0.693 \exp(-\hat{\delta}_{T|T})$.

When $\exp \hat{\delta}_{T|T} + \hat{\gamma}_{T|T} > 0$, the nowcast $\hat{g}_{y,T|T}$ is positive and the estimate of R_t given by (8) is greater than one. So long as $\hat{\gamma}_{T|T}$ is negative, then as $\ell \to \infty$, $\widetilde{R}^e_{\tau,T+\ell|T} \to \exp(\tau\hat{\gamma}_{T|T}) < 1$, and a saturation level for Y appears on the horizon.

We now turn to case where γ_t potentially turns positive in a typically short-lived phase, as a new wave emerges.

3. Reinitialisation

The coronavirus pandemic was characterised by multiple waves punctuated by plateaus. At the beginning of a new wave the growth rate of daily cases, $g_{y,t}$, turns positive. The initial surge may be explosive to the point where the growth is *super-exponential.* In this case, γ_t, the growth rate of g_t (which is the growth rate of cumulative cases) can also turn positive, with no peak in prospect for y_t. Such a phase can be expected to be transient, with γ_t first dropping back toward zero and then becoming negative. When $\gamma_t = 0$, the cumulative growth rate g_t is locally constant, so the implied path for daily cases remains exponential and no finite peak is yet signalled. A future peak in y_t comes into prospect only once γ_t is negative but daily cases are still increasing; at forecast origin T, this corresponds to

$$-\widehat{g}_{T|T} < \widehat{\gamma}_{T|T} < 0,$$

as shown in Section 2.4. If $\widehat{\gamma}_{T|T} \leq -\widehat{g}_{T|T}$, the model instead indicates that daily cases are already at or past their peak.

From the point of view of forecasting an epidemic, a peak must be in prospect even if it can only be expected some way into the future. There is thus a need for a solution to the problem of the estimated $\hat{\gamma}_{t|t}$ rising to positive values as it adapts to the upward surge in y_t, and remaining positive for any protracted period. This upward shift in $\hat{\gamma}_{t|t}$ can be averted by *reinitialising* the $\ln g_t$ series at the start of a new wave. This involves setting the cumulative total of cases Y_t back to zero at, or around, the start of a new wave and setting γ_t to zero so as to impose exponential growth. From the point-of-view of the relationship $g_{y_t} = g_t + \gamma_t$, the re-initialisation effectively shifts "surplus" γ_t emanating from super-exponential growth, into δ_t and therefore into g_t (since $g_t = \exp \delta_t$). Note that, on the date of the re-initialisation, $g_{y,t} = g_t$, since $\gamma_t = 0$, and both $g_{y,t}$ and g_t will be high because a new wave is taking off.

Triggering reinitialisation in real time. Let T denote the current forecast origin and let $\mathcal{F}_T$ denote the information available up to and including date T. Write

$$P_{\gamma,T|T} = \mathrm{Var}(\gamma_T \mid \mathcal{F}_T)$$

for the $(2, 2)$ element of $P_{T|T}$, so that $\sqrt{P_{\gamma,T|T}}$ is the conditional standard deviation of the latent slope γ_T given the information available at T. Define the standardised signal

$$z_{\gamma,T} = \frac{\hat{\gamma}_{T|T}}{\sqrt{P_{\gamma,T|T}}}.$$

In the default implementation, the reinitialisation warning is triggered when

$$z_{\gamma,T} > c_{\text{reinit}}, \qquad c_{\text{reinit}} = 2, \tag{10}$$

that is, when $\hat{\gamma}_{T|T}$ exceeds zero by approximately two conditional standard deviations.

The threshold $c_{\text{reinit}} = 2$ should be interpreted as an operational rule of thumb, motivated by empirical experience with epidemic monitoring, rather than as a fixed statistical decision rule. It provides a simple way to avoid reacting to short-lived fluctuations in the estimated growth dynamics while still detecting sustained departures from the terminal phase of a fitted wave. The procedure is used sequentially, the model is an approximation, and the appropriate threshold depends on the noisiness of the data, the regularity of reporting, and the desired trade-off between early warning and false reinitialisation. In applications, c_{reinit} can therefore be adjusted.

At each forecast origin, the current state is the endpoint of the available sample, so the filtered and smoothed estimates of that state coincide. Hence (10) is an online rule based only on information available at date T. Let d denote the first forecast origin at which (10) holds. The date supplied to `reinit.date` is then the reset date r, defined as the start of the current positive spell in the smoothed slope estimate computed on the sample ending at d:

$$r = \min \left\{ t \leq d : \hat{\gamma}^S_{u|d} > 0 \text{ for all } u = t, \ldots, d \right\}, \tag{11}$$

where $\hat{\gamma}^S_{u|d}$ denotes the smoothed estimate of the slope at date u, using observations only up to and including date d. Thus the trigger date d and the reset date r are distinct: d is the date on which the warning first fires in real time, whereas r is the earlier date at which the cumulative series is reinitialised.

3.1. reinitialising the data series

Let $t = r$ denote the re-initialisation date and let r_0 denote the date at which the cumulative series is set to 0. Then:

$$\begin{aligned} \ln g_t &= \ln y_t - \ln Y_{t-1} & t &= 1, \ldots, r \\ \ln g^r_t &= \ln y_t - \ln Y^r_{t-1} & t &= r+1, \ldots, T \quad (12) \\ Y^r_t &= Y^r_{t-1} + y_t & t &= r, \ldots, T \quad (13) \end{aligned}$$

where Y^r_t denotes the cumulative cases after re-initialisation. We set $Y^r_{r-1} = 0$, so that the growth rate of cumulative cases is available from $t = r+1$ onwards. Note that $Y^r_t = Y_t - Y_{r_0}$. The gap between the two series becomes apparent by writing

$$\ln g^r_t = \ln g_t + \ln \frac{Y_{t-1}}{Y^r_{t-1}} = \ln g_t + \ln \frac{Y_{t-1}}{Y_{t-1} - Y_{r_0}} \quad t = r+1, \ldots, T \tag{14}$$

In the Gauteng illustration in Section 7.1 where we illustrate the working of the program, it can be seen that in contrast to the original $\ln g_t$ series, which continues to increase, the reinitialised $\ln g_t$ series begins to decrease from the reinitialisation date. The reinitialisation enforces the canonical Gompertz curve with the log of growth rate of cumulative cases sloping down.

3.2. Reinitalizing the model

We reinitialize the model by specifying the appropriate prior distribution for the initial states: $\alpha^r_1 \sim N(a^r_1, P^r_1)$ with $a^r_1 = (a^r_{\delta,1}, a^r_{\gamma,1})'$ and P^r_1 defined as follows. Let $\mathcal{F}_t = \ln g_t, \ln g_{t-1}, \ldots, \ln g_1$ and define $a_t = E(\alpha_t|\mathcal{F}_{t-1})$, $a_{t|t} = E(\alpha_t|\mathcal{F}_t)$, $P_t = Var(\alpha_t|\mathcal{F}_{t-1})$, $P_{t|t} = Var(\alpha_t|\mathcal{F}_t)$. Then,

$$
\begin{aligned}
a^r_{\delta,1} &= a_{\delta,r+1} + \ln(Y_r/y_r) \\
a^r_{\gamma,1} &= 0 \\
P^r_1 &= P_{r+1},
\end{aligned}
$$

where $a_{\delta,r+1}$ and P_{r+1} are obtained from the non-reinitialised model estimated over $t = 1, \ldots, r$ via the usual Kalman filter recursions. Adding $\ln(Y_r/y_r)$ to $a_{\delta,r+1}$ corrects for the shift down in the log cumulative cases caused by reinitialising the cumulative case series. Setting $a^r_{\gamma,1} = 0$ ensures the model starts off with exponential, rather than super-exponential, growth.

We reinitialize the model through the priors in this way rather than simply re-estimating the model from scratch for two reasons. First, it allows us to impose a proper (rather than diffuse) prior centered on zero for γ, so that the starting point is exponential growth. Second, it enables us to make use of data from before the reinitialisation date. One needs a reasonable sample size for the estimated model and forecasts to be reliable, but if a new wave is taking off, forecasts need to be generated quickly. This was particularly true with the emergence of the Omicron variant in December 2021, which caused an explosive increase in infection over a short period of time.

We do not re-estimate the σ^2_ε or σ^2_ζ parameters in the reinitialised model. Rather, we use the values estimated in the original model over $t = 1, \ldots, r$. The first observation in the reinitialised log-growth series is at calendar time $r + 1$, since $Y^r_{r-1} = 0$ and hence $\log g^r_r$ is not defined. The prior a^r_1 therefore refers to the state used to predict the first reinitialised observation, $\log g^r_{r+1}$. By construction,

$$
v^r_{r+1} = \log g^r_{r+1} - a^r_{\delta,1} = \log g_{r+1} - a_{\delta,r+1} = v_{r+1},
$$

so the first one-step-ahead prediction error after reinitialisation is the same as in the original model. From calendar time $r + 2$ onwards, the prediction errors generally diverge, because the reinitialised model sets $a^r_{\gamma,1} = 0$, whereas the original model carries forward $a_{\gamma,r+1}$.

The reinitialisation procedure is very similar in the case where we have seasonal terms. If we let $\alpha_{s,t}$ be the vector of seasonal states and maintain an analogous notation to that above, the prior mean of the seasonal components in the reinitialised model is

$$
a^r_{s,1} = a_{s,r+1}.
$$

The prior variance of α^r_1 remains P_{r+1} but P_{r+1} is re-defined to include the seasonal term in the manner described in the Appendix.

4. The Leading Indicator Model

In some applications, leading indicators—time series that systematically move ahead of the target variable by a certain number of periods—are available. Extending the dynamic Gompertz framework to incorporate such information can improve short-horizon forecasting. The basic idea is that two related series share a common underlying growth component, but one series leads the other by k periods. Examples include COVID-19 cases leading hospitalisations within a region, or epidemic outcomes in a country experiencing an earlier outbreak leading those in a country that lags behind.

Let $g_{1,t-k}$ denote the growth rate of the leading series and $g_{2,t}$ the growth rate of the target series. The leading-indicator model is

$$\ln g_{1,t-k} = \delta_t + \psi_t + \varepsilon_{1t}, \qquad t = k+1, \ldots, T+k, \tag{15}$$
$$\ln g_{2,t} = \delta_t + \varepsilon_{2t}, \qquad t = k+1, \ldots, T, \tag{16}$$

where:

- $\ln g_{1,t-k}$ is the log growth rate of the leading series observed k periods earlier than the target;
- $\ln g_{2,t}$ is the log growth rate of the target series at time t;
- δ_t is the common time-varying component, evolving as in the dynamic Gompertz model in Section 2;
- ψ_t is a time-varying deviation in the leading equation;
- ε_{1t} and ε_{2t} are mutually independent observation disturbances; and
- k is the lead–lag parameter.

The lag k is an alignment parameter rather than a distributed-lag coefficient. The leading series does not enter the target equation as an exogenous regressor; instead, after shifting by k, the lead and target growth rates are treated as two noisy measurements of a common latent growth component. Thus the current **tsgc** implementation uses a single scalar lag k for each fitted leading-indicator model. When the timing relationship is uncertain, we treat k as a model-selection parameter and compare separate single-lag specifications over a candidate grid using rolling-origin forecast evaluation. This is the role of `cross_val()` in the empirical illustration below. A model that combines several lagged values of the observed leading series in one target equation would be a different distributed-lag or transfer-function specification, and is not implemented here.

State-space representation. It is useful to write the leading-indicator model explicitly as a bivariate state-space system. Let T denote the final date for which the target series is observed and let T_1 denote the final date for which the leading series is observed, both measured on the same calendar time scale. After aligning the leading series by the lag k, define

$$z_{1t} = \log g_{1,t-k}, \qquad z_{2t} = \log g_{2,t}.$$

Thus z_{1t} is observed whenever the leading series is observed at date $t-k$, while z_{2t} is observed whenever the target series is observed at date t.

For the baseline leading-indicator specification, the state vector is

$$\alpha_t = \begin{pmatrix} \delta_t \\ \gamma_t \\ \psi_t \end{pmatrix},$$

where δ_t is the common dynamic Gompertz component, γ_t is its local slope, ψ_t is the deviation in the leading equation. The observation equation is

$$z_t = \begin{pmatrix} z_{1t} \\ z_{2t} \end{pmatrix} = Z\alpha_t + \varepsilon_t, \qquad Z = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \tag{17}$$

with

$$\varepsilon_t = \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} \sim NID(0, H), \qquad H = \begin{pmatrix} \sigma^2_{\varepsilon 1} & 0 \\ 0 & \sigma^2_{\varepsilon 2} \end{pmatrix}.$$

The transition equation is

$$\alpha_{t+1} = T_{\mathrm{LI}}\alpha_t + R_{\mathrm{LI}}\eta_t, \tag{18}$$

where

$$T_{\mathrm{LI}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad R_{\mathrm{LI}} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and

$$\eta_t = \begin{pmatrix} \zeta_t \\ \xi_t \end{pmatrix} \sim NID(0, Q), \qquad Q = \begin{pmatrix} \sigma^2_{\zeta} & 0 \\ 0 & \sigma^2_{\xi} \end{pmatrix}.$$

Equivalently, the state equations are

$$\delta_{t+1} = \delta_t + \gamma_t, \qquad \gamma_{t+1} = \gamma_t + \zeta_t, \qquad \psi_{t+1} = \psi_t + \xi_t.$$

We define the signal-to-noise ratio $q_\xi = \sigma^2_\xi/\sigma^2_{\varepsilon 2}$. All observation and state disturbances are assumed mutually independent. Estimation is carried out by maximum likelihood using the Kalman filter, with diffuse initialisation for the unknown initial states.

Forecasting with partially observed lead values. The aligned representation in (17) also clarifies the out-of-sample mechanics. Suppose the target series is observed through date T, the leading series through date T_1, and an h-step-ahead target forecast is required. At target forecast horizon ℓ, the aligned lead observation is $z_{1,T+\ell} = \log g_{1,T+\ell-k}$. This value is available to the forecaster if and only if

$$T + \ell - k \leq T_1.$$

Hence the number of target forecast steps that can condition on observed lead values is

$$L_T = \min\{h, \max(0, T_1 - T + k)\}.$$

When $T_1 = T$, so that the lead and target series are observed through the same calendar date, this reduces to $L_T = \min\{h, k\}$.

The Kalman recursion treats unavailable future lead values as missing observations. For $t = T+1, \ldots, T+h$, let $\mathcal{I}_t$ denote the set of observed rows of z_t. In the forecast period the target row is missing by construction, while the lead row is included only when z_{1t} is available. The one-step prediction equations are

$$a_{t|t-1} = T_{\mathrm{LI}} a_{t-1|t-1}, \qquad P_{t|t-1} = T_{\mathrm{LI}} P_{t-1|t-1} T'_{\mathrm{LI}} + R_{\mathrm{LI}} Q R'_{\mathrm{LI}}.$$

If $\mathcal{I}_t$ is non-empty, the available observation rows are used to update the state:

$$v_t = z_{\mathcal{I}_t,t} - Z_{\mathcal{I}_t} a_{t|t-1}, \qquad F_t = Z_{\mathcal{I}_t} P_{t|t-1} Z'_{\mathcal{I}_t} + H_{\mathcal{I}_t},$$

$$a_{t|t} = a_{t|t-1} + P_{t|t-1} Z'_{\mathcal{I}_t} F_t^{-1} v_t,$$
$$P_{t|t} = P_{t|t-1} - P_{t|t-1} Z'_{\mathcal{I}_t} F_t^{-1} Z_{\mathcal{I}_t} P_{t|t-1}.$$

If $\mathcal{I}_t$ is empty, no update is made and

$$a_{t|t} = a_{t|t-1}, \qquad P_{t|t} = P_{t|t-1}.$$

The forecast of the target log growth rate is then

$$\widehat{\log g}_{2,t|T} = Z_2 a_{t|t}, \qquad Z_2 = \begin{pmatrix} 1 & 0 & 1 \end{pmatrix},$$

where $a_{t|t}$ conditions on the lead observation at date t if that observation is available. The corresponding level forecasts are obtained recursively from

$$\widehat{Y}_{2,t|T} = \widehat{Y}_{2,t-1|T} \left(1 + \exp\{\widehat{\log g}_{2,t|T}\}\right),$$

with initial value $\widehat{Y}_{2,T|T} = Y_{2,T}$, and

$$\widehat{y}_{2,t|T} = \widehat{Y}_{2,t-1|T} \exp\{\widehat{\log g}_{2,t|T}\}.$$

Thus the method does not require artificial imputation of missing future lead values. Known lead values are used as observations; unavailable lead values are skipped by the Kalman filter.

Seasonality and regressors. Seasonal effects and explanatory variables enter through the observation equation. With equation-specific regressors, the leading-indicator observation equations become

$$z_{1t} = \delta_t + \psi_t + x'_{1,t-k}\beta_1 + \varepsilon_{1t}, \tag{19}$$
$$z_{2t} = \delta_t + x'_{2,t}\beta_2 + \varepsilon_{2t}, \tag{20}$$

where $x_{1,t-k}$ enters only the lead equation and $x_{2,t}$ enters only the target equation. Equivalently, the state vector can be augmented as

$$\alpha_t^x = \begin{pmatrix} \delta_t \\ \gamma_t \\ \psi_t \\ \beta_1 \\ \beta_2 \end{pmatrix},$$

with time-varying observation matrix

$$Z_t^x = \begin{pmatrix} 1 & 0 & 1 & x'_{1,t-k} & 0 \\ 1 & 0 & 0 & 0 & x'_{2,t} \end{pmatrix}.$$

The regression coefficients are static states, or equivalently fixed coefficients, with zero transition variance. Seasonal components are appended to the state vector in the same way as in the univariate model, with the relevant seasonal loading added to the corresponding observation row.

In the package implementation, `supply_xpred.new()` supplies the future design matrices $x_{1,t-k}$ and $x_{2,t}$ used in (19)-(20). Calling `supply_xpred.new(object, X, idx = "lead")` updates the future covariate path for the lead equation; calling `supply_xpred.new(object, X, idx = "targ")` updates the future covariate path for the target equation. The estimated coefficients are not re-estimated by this step. The function only supplies the future or assumed covariate values that enter the observation matrix during forecasting. If future covariates are not observed in real time, they must be supplied as external forecasts or scenario paths; otherwise the corresponding regressor-augmented forecast is not fully specified.

The usefulness of the leading-indicator model depends on the empirical stability of the relationship between the leading and target series over the forecast horizon. The specification is motivated by balanced growth in an empirical sense: after aligning the two series by the lag k, they are assumed to share a common latent growth component, with additional departures absorbed by ψ_t. No explicit test for cointegration is currently included. Structural breaks—for example, due to major policy changes, changes in testing or reporting, or shifts in public behaviour—may weaken forecast performance. Likewise, the lag k is treated as fixed within a given specification, so lag choice should be regarded as a model-selection problem to be examined empirically, for example over a candidate lag grid.

Later empirical illustrations show how this framework can be used with weekly seasonality and optional regressors in an England cases–hospitalisations application, and how it can be compared with the univariate dynamic Gompertz model in a UK–Italy application where lag choice is assessed by rolling cross-validation.

5. Forecast accuracy

Forecast evaluation for time series must preserve the temporal ordering of the data. We therefore use rolling-origin (walk-forward) procedures that mimic real-time forecasting rather than random sample splitting. When competing specifications are compared, the forecast horizon and holdout design are kept fixed across models so that differences in forecast accuracy are attributable to the models themselves.

For a fixed model specification and forecast horizon h, out-of-sample performance is assessed by a holdout construction. Let T_0 denote the final date in the truncated estimation sample, chosen so that observations are available for the holdout period $T_0+1, \ldots, T_0+h$. The model is estimated using observations up to T_0, and an h-period forecast path,

$$\left\{\widehat{y}_{T_0+\ell|T_0} : \ell = 1, \ldots, h\right\},$$

is produced. This forecast path is then compared with the withheld observations,

$$\left\{y_{T_0+\ell} : \ell = 1, \ldots, h\right\}.$$

This provides a direct measure of short-horizon forecast accuracy under a design that matches the practical forecasting problem.

For the leading-indicator model, the lag k is treated as a model-selection parameter. This procedure fits separate single-lag leading-indicator models; it does not estimate a distributed-lag regression with several lagged values of the lead series entering the target equation simultaneously. We therefore use rolling cross-validation over a grid of candidate lags, $k \in \mathcal{K}$. For each candidate lag and each forecast origin, the model is re-estimated, forecasts up to h steps ahead are generated, and the corresponding forecast errors over horizons $\ell = 1, \ldots, h$ are recorded. The lag can then be selected by summarizing the errors across forecast origins, for example by their mean or median under a chosen accuracy criterion.

We foreground the symmetric Mean Absolute Percentage Error (sMAPE), reported in percentage points, and also report the Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) for context. Let T_0 denote the forecast origin, so that the model is estimated using observations up to T_0, and let

$$\left\{\widehat{y}_{T_0+\ell|T_0} : \ell = 1, \ldots, h\right\}$$

denote the corresponding h-period forecast path. The withheld observations are

$$\{y_{T_0+\ell} : \ell = 1, \ldots, h\}.$$

The accuracy measures are then defined as

$$\text{sMAPE} = \frac{100}{h} \sum_{\ell=1}^{h} \frac{\left|y_{T_0+\ell} - \widehat{y}_{T_0+\ell|T_0}\right|}{\left(\left|y_{T_0+\ell}\right| + \left|\widehat{y}_{T_0+\ell|T_0}\right|\right)/2},$$

$$\text{MAPE} = \frac{100}{h} \sum_{\ell=1}^{h} \left|\frac{y_{T_0+\ell} - \widehat{y}_{T_0+\ell|T_0}}{y_{T_0+\ell}}\right|,$$

$$\text{MAE} = \frac{1}{h} \sum_{\ell=1}^{h} \left|y_{T_0+\ell} - \widehat{y}_{T_0+\ell|T_0}\right|,$$

$$\text{RMSE} = \sqrt{\frac{1}{h} \sum_{\ell=1}^{h} \left(y_{T_0+\ell} - \widehat{y}_{T_0+\ell|T_0}\right)^2}.$$

Lower values indicate better forecast performance. Because MAPE can be sensitive when observed values are very small, sMAPE is used as the primary percentage-based comparison measure in the empirical illustrations below. Taken together, the holdout exercise and the rolling cross-validation procedure provide a practical workflow for comparing model classes and, in the leading-indicator case, for selecting the lag structure empirically.

Implementation in the replication. The replication implements the holdout and rolling-origin procedures described above. For leading-indicator models, `cross_val()` evaluates candidate lags over a user-specified lag grid and records forecast errors over successive forecast origins. The empirical illustrations reported later in the paper summarize forecast performance using sMAPE, MAPE, MAE, and RMSE.

6. Functionality of tsgc

The package is organised around two main model classes, `SSModelDynamicGompertz` and `SSModelLeadingIndicator`, which implement the models in (2)–(4) and (15)–(16). Both classes construct the relevant state space model and share a common estimation workflow via `estimate()`. Estimation returns a `FilterResults` or `FilterResultsLI` object, extending the underlying **KFAS** `KFS` output with a date index and forecasting methods.

For the dynamic Gompertz model, the required input is a cumulative series `Y`. Optional arguments allow the user to specify the start date and end date of the estimation window through `start.date` and `end.date` if not the whole dataset is to be used, fix the signal-to-noise ratio q_ζ, include AR(1) extension through `ar1`, include seasonality through `sea.period`, include exogenous predictors with `xpred` and reinitialize the model at a new-wave date using `reinit.date`. Here `reinit.date` is the reset date r defined in (11), not the date on which the warning is first triggered. The trigger date is the first forecast origin d for which the endpoint condition in (10) holds. Reinitialisation may either carry forward information from an earlier fit via `original.results`, or treat the post-`reinit.date` segment as a fresh wave by setting `use.presample.info = FALSE`. The resulting `FilterResults` object provides filtered or smoothed estimates of growth rates through `get_growth_y()` and `get_gy_ci()`, together with level and state forecasts through `predict_level()` and `predict_all()`.

For the leading-indicator model, `SSModelLeadingIndicator` takes the joint lead–target series, the lag structure, and optional start date, end date, signal-to-noise ratio, seasonal and regressor inputs. Exogenous predictors can be included in the lead and target equations, and when such regressors are used, future covariate paths are supplied through `supply_xpred.new()` before forecasting. Estimation returns a `FilterResultsLI` object, which supports the same broad forecasting workflow as in the univariate case.

A small set of helper functions supports the main empirical workflow. `plot_log_forecast()` plots forecasts of $\ln(g_t)$, `plot_forecast()` plots forecasts of the incidence variable together with the state-implied forecast bands described in Section 2.4, and `plot_holdout()` compares forecasts with realised outcomes over an evaluation sample. Diagnostic plots of the filtered or smoothed growth components and their pointwise uncertainty bands are available through `plot_gy_components()` and `plot_gy_ci()`. Reinitialisation of a data frame at a given date is handled by `reinitialise_dataframe()` and rolling lag selection by `cross_val()`.

The same workflow applies across daily, weekly, monthly, or quarterly data. What changes with the sampling frequency is the interpretation of the time index, forecast horizon, lag length, and seasonal period.

Further details are given in the empirical illustrations in the next section.

7. Illustration of the tsgc package

This section presents three worked examples showing how **tsgc** is used in practice. The first uses Gauteng COVID-19 cases to illustrate the core dynamic Gompertz workflow and reinitialisation across waves. The second uses England daily cases and hospitalisations to illustrate the leading-indicator model, first with weekly seasonality only and then with additional regressors. The third compares a UK-only dynamic Gompertz model with an Italy *to* UK leading-indicator model and uses rolling cross-validation to assess lag choice.

Throughout the empirical illustrations, the argument `confidence.level` controls the width of the visual uncertainty bands. The interpretation of these bands depends on the plotted quantity. For $\ln(g_t)$, the bands are derived from the Gaussian state-space prediction or filtering distribution. For new cases y_t, the bands are state-implied forecast bands obtained by propagating bands for the latent log growth rate through the nonlinear incidence recursion;

they are not formal prediction intervals for realised daily counts. For R_t, the bands are approximate pointwise uncertainty bands for model-implied reproduction-number summaries, conditional on the chosen generation interval.

The current development version can be installed from GitHub; the CRAN release can be installed in the usual way.

```
remotes::install_github("edwintang903/tsgc")
library(tsgc)
```

7.1. Gauteng: dynamic Gompertz workflow and reinitialisation

The package includes example COVID-19 data for Gauteng province in South Africa. The series is stored in cumulative form as an `xts` object with a date index.

```
data(gauteng, package = "tsgc")
```

Figure 1 plots daily new cases and their centered 7-day moving average for Gauteng between 10 March 2020 and 5 January 2022. The sequence of waves makes the series a useful illustration of both the baseline dynamic Gompertz workflow and the reinitialisation procedure.

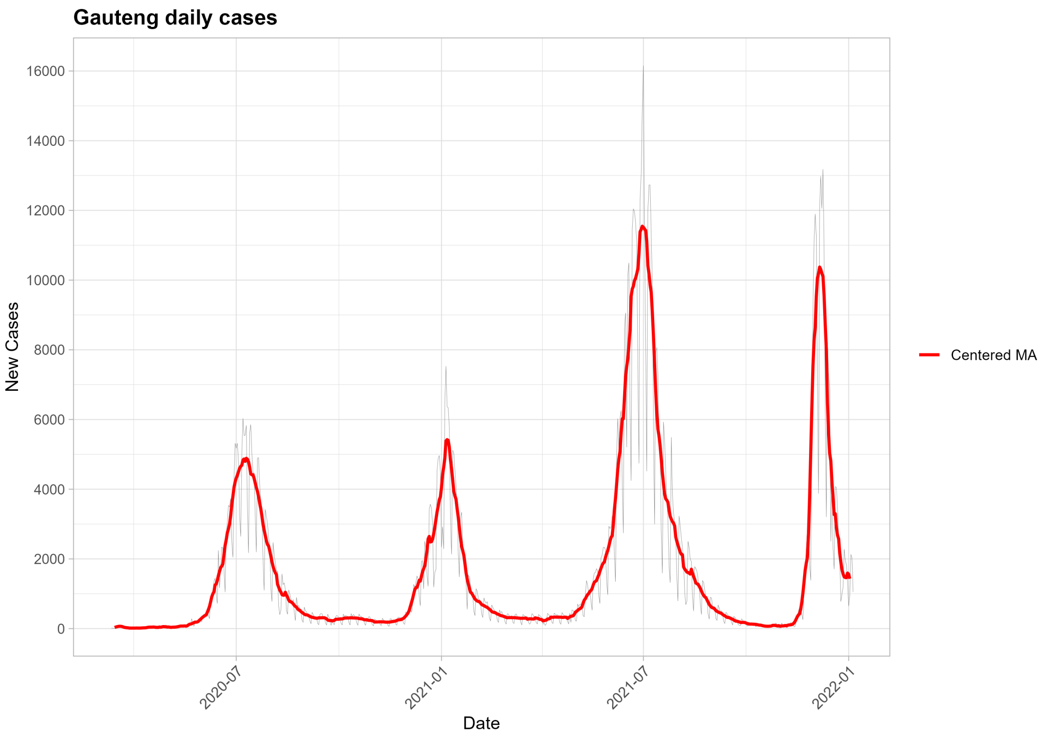

Figure 1: New cases and their centered 7-day moving average for Gauteng province in South Africa between 10 March 2020 and 5 January 2022.

Setting up the forecasting exercise

We begin with the early phase of Gauteng's third wave. Suppose we only have access to data up to 3 May 2021. For our model, we define our estimation window to be from 1 February 2021 to 3 May 2021 and the signal-to-noise ratio is set to `q = 0.005`. In the output, we want a forecast horizon of 14 days, confidence bands corresponding to a confidence level of 0.68 and the plotting window showing the dataset 30 days before the end of estimation.

```
cumulative_cases <- gauteng[, 1]
n.forecasts <- 14
q.default   <- 0.005
plt.length  <- 30
est.start.1 <- as.Date("2021-02-01")
est.end.1   <- as.Date("2021-05-03")
CONF_LEVEL   <- 0.68
```

Estimation and baseline forecasts

The signal-to-noise ratio can either be estimated or fixed. We show both steps, but in the remainder of the example we work with the fixed value `q = 0.005`, which provides a useful balance between smoothness and responsiveness in coronavirus applications.

```
# Diffuse prior (free q)
model_free <- SSModelDynamicGompertz(Y = cumulative_cases,
    start.date = est.start.1, end.date = est.end.1)
res_free <- estimate(model_free)

# Fixed q
model_q <- SSModelDynamicGompertz(Y = cumulative_cases, q = q.default,
    start.date = est.start.1, end.date = est.end.1)
res_q <- estimate(model_q)
```

The fitted object `res_q` can then be used to plot forecasts of $\ln(g_t)$ and forecasts of new cases. These two forecast plots can be found in Figures 2 and 3 respectively.

```
plot_log_forecast(
  res_q, Y = cumulative_cases, n.ahead = n.forecasts,
  plt.start.date = tail_date_minus(res_q$index, plt.length),
  title = "Forecast of log growth rate of cases 14-day (Gauteng)")

plot_forecast(
  res_q, n.ahead = n.forecasts, confidence.level = CONF_LEVEL,
  plt.start.date = tail_date_minus(res_q$index, plt.length),
  title = "Forecast of new cases 14-days (Gauteng)",
  series.name = "Cases")
```

In practice, to evaluate the model's empirical accuracy, we can perform a holdout analysis. We shift the estimation end-date back by 14 days (n.forecasts) to simulate a scenario where the most recent two weeks of data are unknown. This retrospective forecast provides a benchmark for the model's predictive power, as the resulting estimates can be directly validated against the actual observed case counts that were excluded during the fitting process.

```
est.end.holdout <- est.end.1 - n.forecasts
model_q_holdout <- SSModelDynamicGompertz(
  Y = cumulative_cases,
  q = q.default,
  start.date = est.start.1,
  end.date   = est.end.holdout)
res_q_holdout <- estimate(model_q_holdout)
```

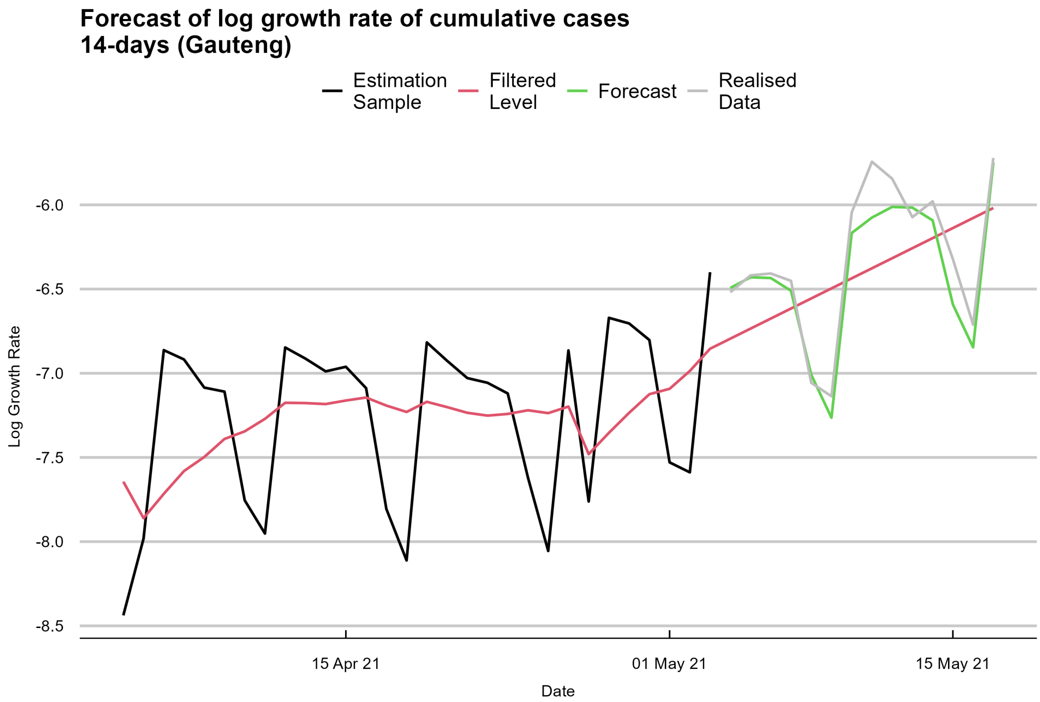

Figure 2: Fourteen-day forecast of $\ln(g_t)$ from 4 May 2021 for Gauteng province in South Africa.

In this early third-wave window the model performs reasonably well over the first week of the holdout period, as seen from Figure 4. The 14-day holdout MAPE is 27%, while over the first seven days it is materially lower. The main deviation is associated with the unusually low case count reported on 27 April 2021, a public holiday in South Africa.

Forecast tables can be exported once a results directory has been specified.

```
base_path   <- getwd()
results_dir <- file.path(base_path, "results")
tables_dir  <- file.path(results_dir, "Tables")

write_results(
    res                  = res_q,
    res.dir              = tables_dir,
    n.ahead              = n.forecasts,
```

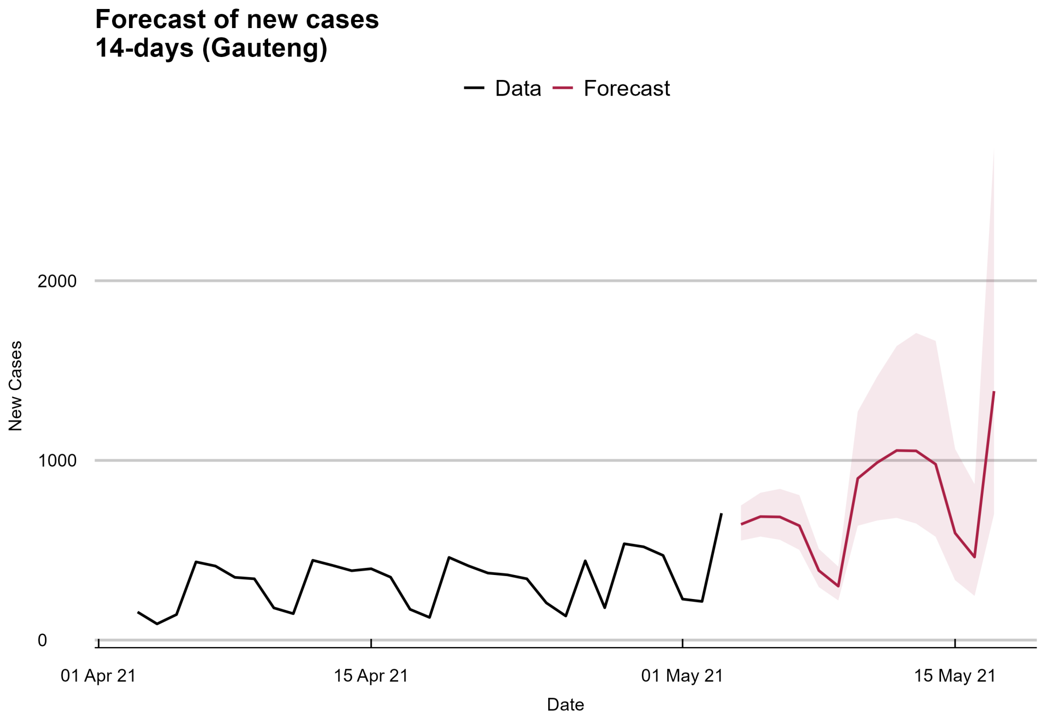

Figure 3: Fourteen-day forecast of new cases from 4 May 2021 for Gauteng province in South Africa. The shaded region is a state-implied forecast band obtained by propagating uncertainty in the latent log growth rate.

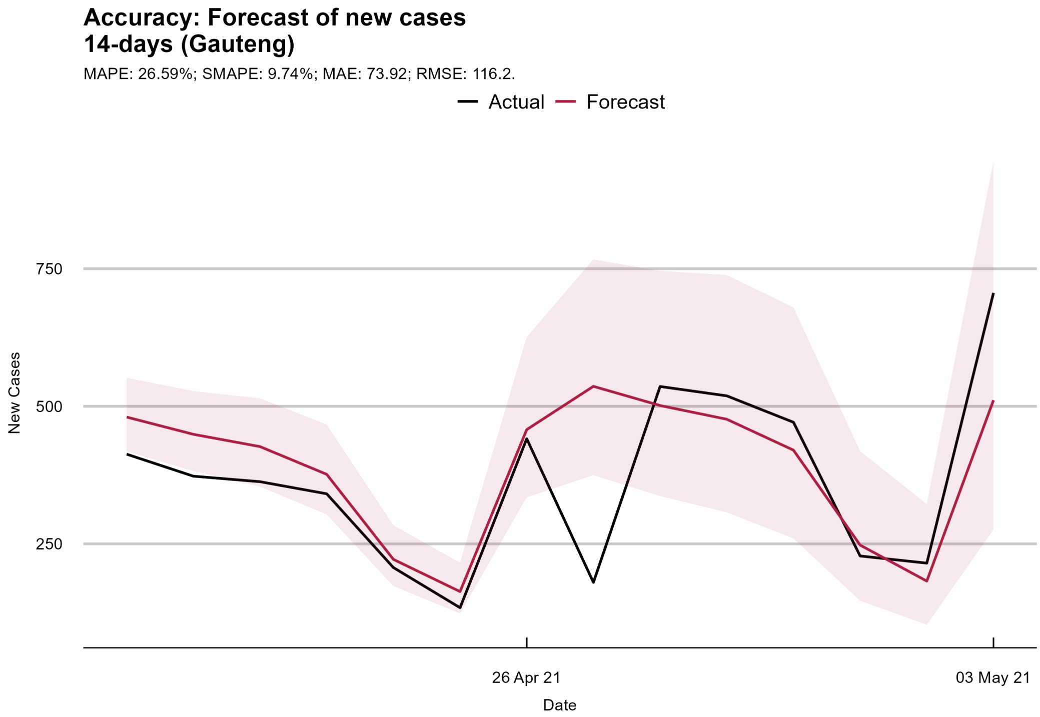

Figure 4: Accuracy of the fourteen-day forecast of new cases from 20 April 2021 for Gauteng province in South Africa.

```
    confidence.level  = CONF_LEVEL,
    prefix            = "gauteng_gomp_q005_")
```

Reproduction numbers

As discussed in Section 2.5, the filtered growth-rate estimates can be translated into instantaneous reproduction numbers. Using a generation interval of four days, the corresponding estimates are obtained as follows. The output plot is presented in Figure 5.

```
gen_int <- 4
ndays   <- 7
r.t <- estimate_r0(res_q, gen_int, ndays)

# Show plot
estimate_r0(res_q, gen_int, ndays, show_plot = TRUE,
            title = "Gauteng Reproduction numbers")
```

These estimates provide a compact summary of whether transmission is likely to be expanding or contracting around the forecast origin.

Reinitialisation across waves

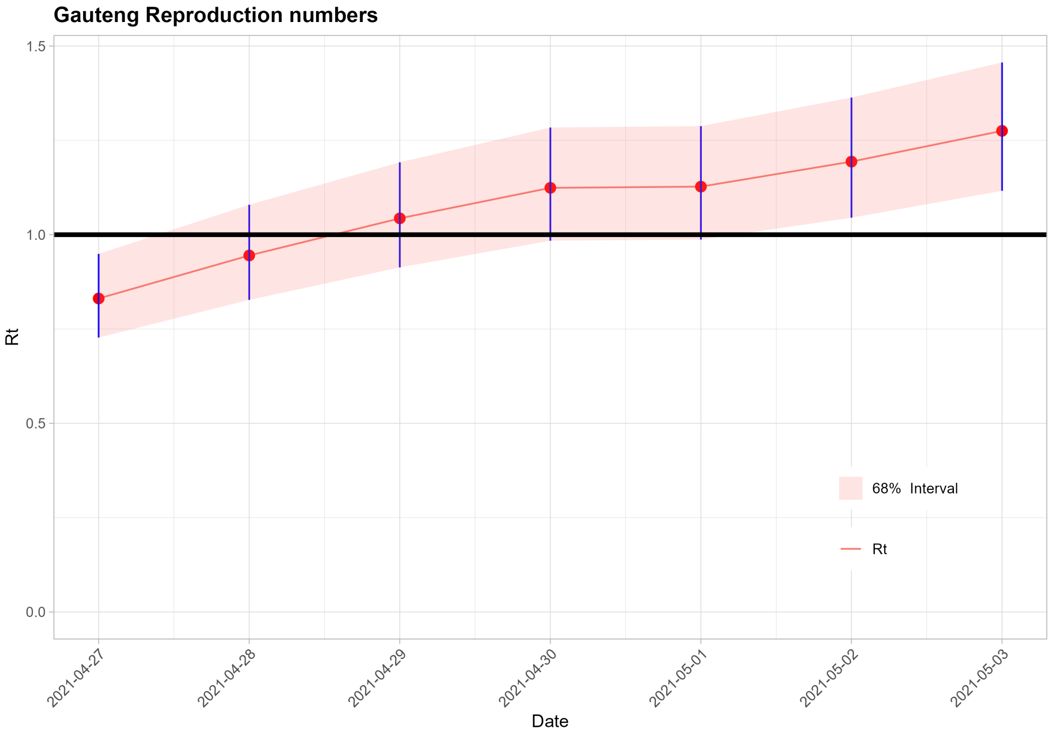

Figure 5: Model-implied reproduction numbers for the 7-day period to 3 May 2021 for Gauteng province in South Africa. Bands are approximate pointwise uncertainty bands conditional on a four-day generation interval.

To illustrate reinitialisation, we extend the estimation window to 25 June 2021, by which point the third wave is close to its peak. Illustrated in Figure 6, applying the real-time rule in (10) to the sample ending on 25 June 2021 shows that the endpoint smoothed estimate of the slope parameter γ_t exceeds zero by more than two standard errors. The reset date is then backdated according to (11). In this example, the resulting **reinit.date** is 21 April 2021.

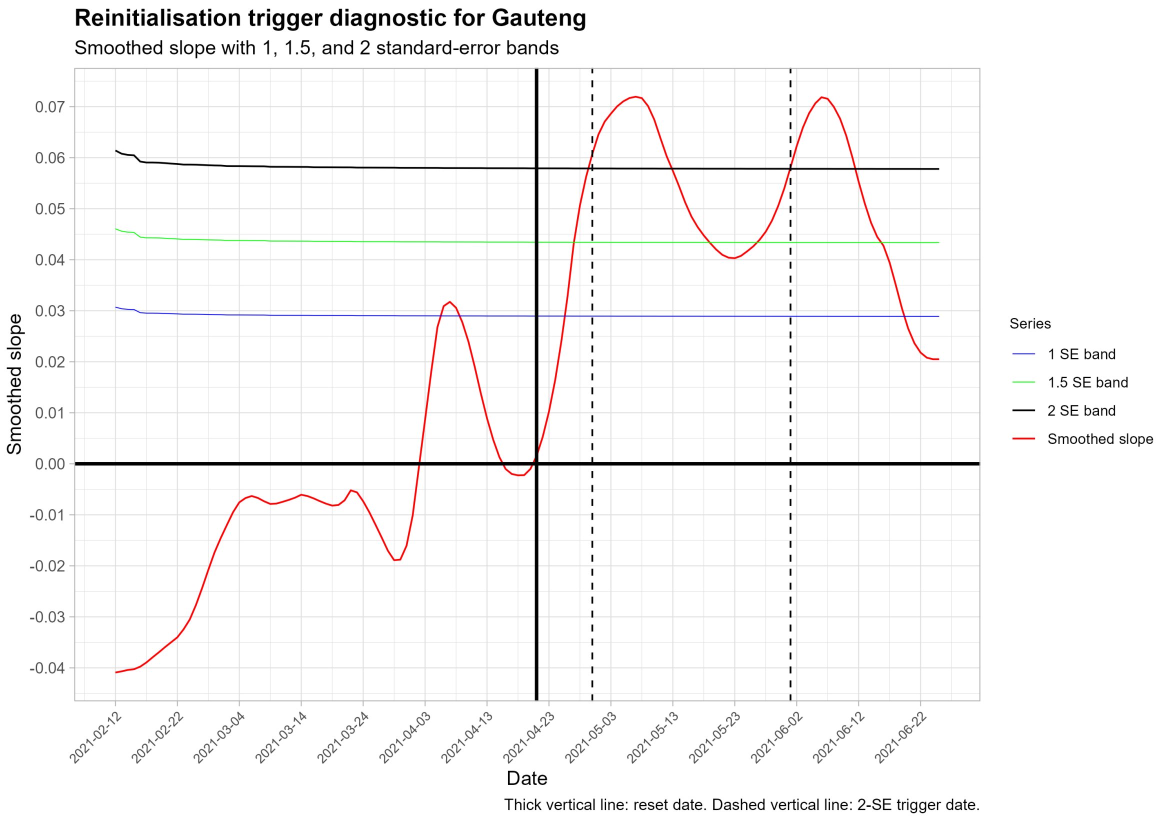

Figure 6: Reinitialisation rule for the sample ending on 25 June 2021. The endpoint trigger is satisfied when $\hat{\gamma}_{T|T}$ exceeds zero by two standard errors. The reset date supplied to **reinit.date** is then backdated to 21 April 2021, the start of the current positive spell in the smoothed slope estimate computed on that same sample.

```
est.end.2 <- as.Date("2021-06-25")
reinit.date <- as.Date("2021-04-21")

model_reinit <- SSModelDynamicGompertz(
        Y = cumulative_cases, q = q.default,
        start.date = est.start.1, end.date = est.end.2,
        reinit.date = reinit.date)
res_reinit <- estimate(model_reinit)
```

The reinitialised model is handled exactly like the baseline model: we can plot forecasts of $\ln(g_t)$, forecasts of new cases, and the corresponding holdout comparison.

```
plot_log_forecast(
  res_reinit, Y = cumulative_cases, n.ahead = n.forecasts,
  plt.start.date = tail_date_minus(res_reinit$index, plt.length),
  title = "Forecast of log growth rate of cases after reinitialisation")

plot_forecast(
  res_reinit, n.ahead = n.forecasts, confidence.level = CONF_LEVEL,
  plt.start.date = tail_date_minus(res_reinit$index, plt.length),
  title = "Forecast of new cases after reinitialisation",
  series.name = "Cases")

plot_holdout(
  res_reinit, Y = cumulative_cases, n.ahead = n.forecasts,
  confidence.level = CONF_LEVEL,
  title = "Accuracy: Forecast of new cases with reinitialisation",
  series.name = "Cases")
```

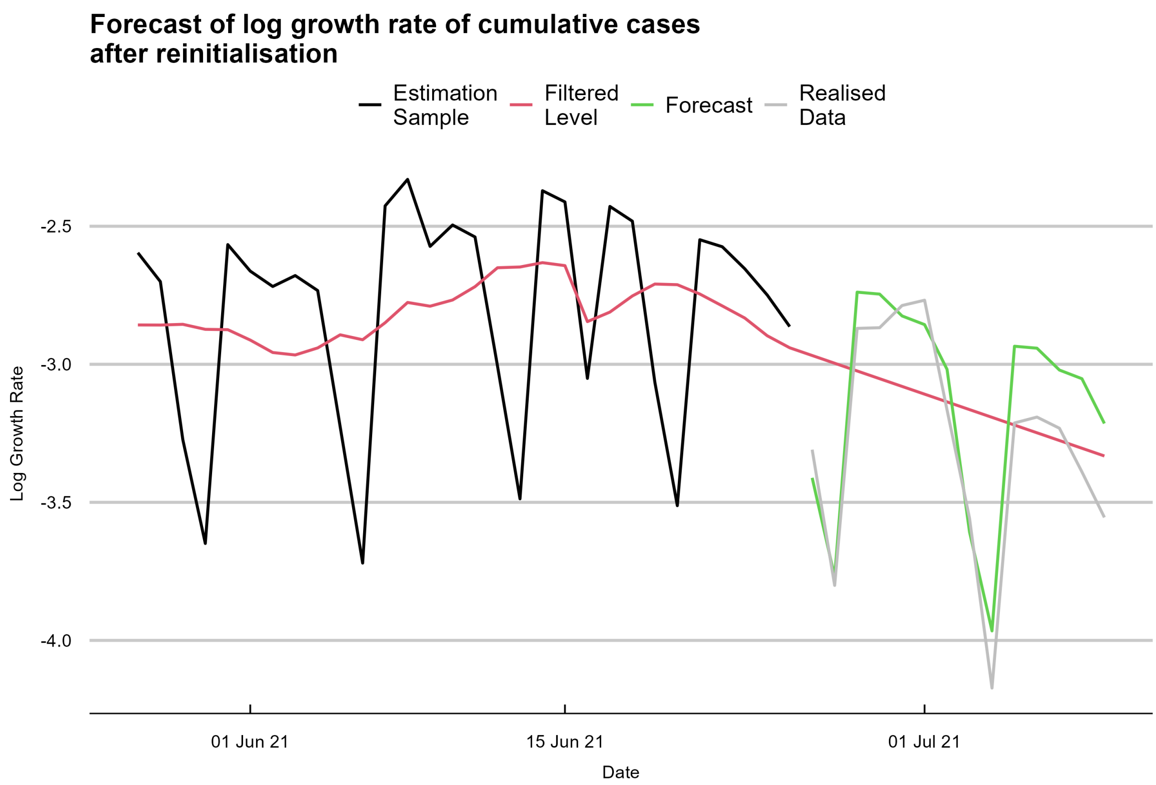

Figure 7: Forecast of $\ln(g_t)$ after reinitialisation.

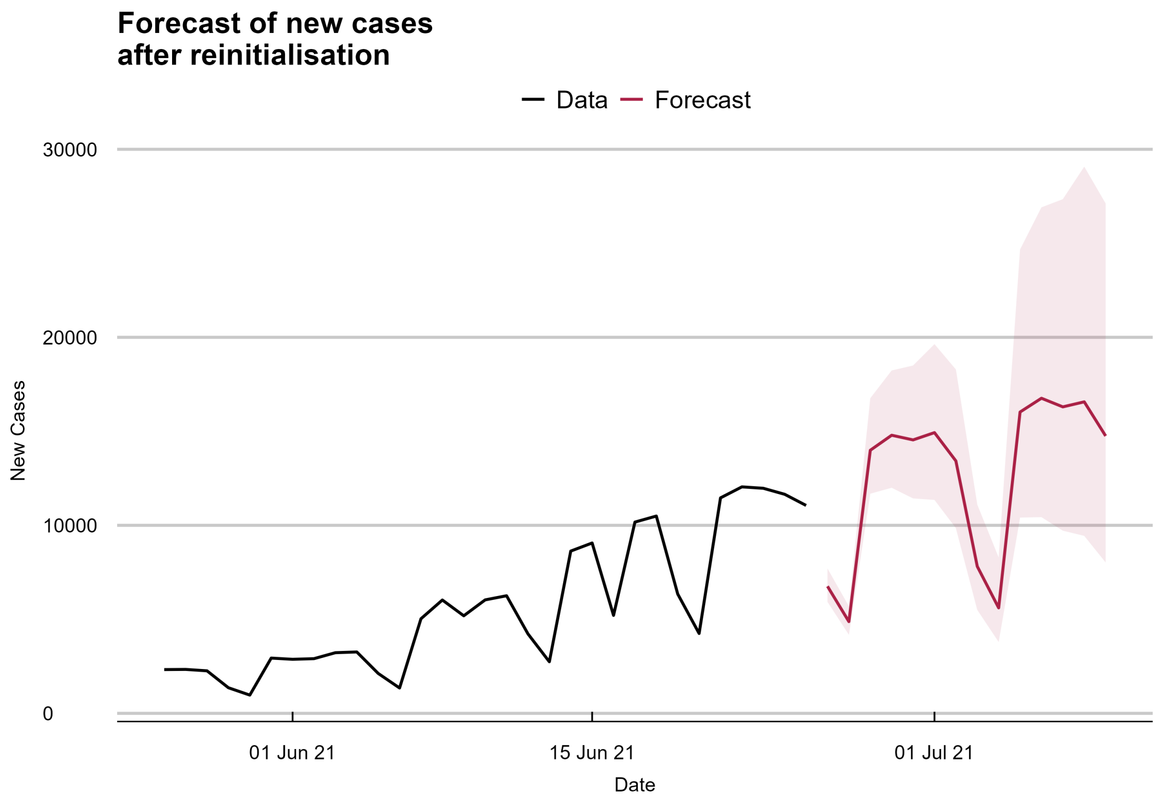

Figure 8: Forecast of new cases after reinitialisation. The shaded region is a state-implied forecast band.

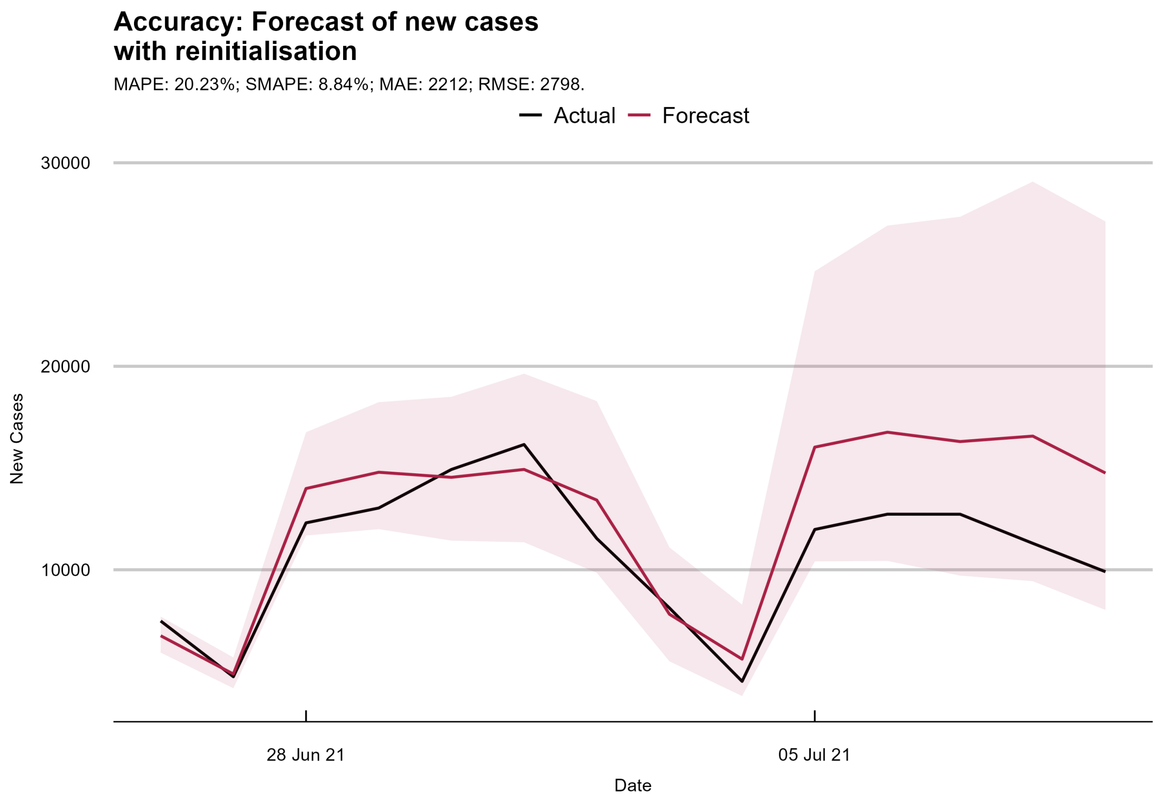

Figure 9: Forecast accuracy of the reinitialised model over the holdout period: 14 days from 26 June 2021.

For the 14 days from 26 June 2021, when the peak is imminent and the trajectory is difficult to forecast accurately, reinitialisation substantially improves performance. In the supplied replication results, the 14-day holdout MAPE falls from 41.9% without reinitialisation to 20.2% with reinitialisation. Over the first seven days, the corresponding figures fall from 15.2% to 9.5%. The Gauteng example therefore illustrates both the baseline dynamic Gompertz workflow and the practical value of reinitialisation when a new wave is underway.

7.2. England: leading-indicator workflow

We now turn to the leading-indicator model. This example uses daily England COVID-19 cases as the lead series and daily hospitalisations as the target series. The aim is to illustrate the practical workflow for estimating the bivariate state-space specification, producing short-horizon forecasts, and then extending the same model with additional regressors.

Baseline model with weekly seasonality

A preliminary plot of the two series on the log scale is useful for inspecting comovement before formal estimation.

```
data(england, package = "tsgc")
eng <- tsgc::england[, 1:2]

# Quick plot
mod2 <- SSModelLeadingIndicator(eng, n.lag = 4)
plot(mod2, title = "Daily COVID cases and Hospitalisations\n(England)",
  series.name.lead = "Cases", series.name.target = "Hospitalisations",
  take.log = TRUE
)
```

For the leading-indicator model, `SSModelLeadingIndicator` constructs an aligned bivariate state-space system from the lead and target series. The lead equation is shifted by the specified lag `n.lag`, so that observed lead values can enter the target forecast window. During forecasting, available future-aligned lead observations are used as observations in the Kalman recursion, while unavailable lead values are treated as missing. Thus, if the target forecast horizon exceeds the number of available aligned lead observations, the remaining target forecasts are generated by propagation of the state equations rather than by imputing the leading series.

Exogenous predictors may be included separately in the lead and target equations through `xpred_lead` and `xpred_targ`. Future or assumed covariate paths are supplied after estimation using `supply_xpred.new()`. The argument `idx = "lead"` supplies the future design matrix for the lead equation, while `idx = "targ"` supplies the future design matrix for the target equation. This step does not re-estimate the model; it provides the future covariate values used by the forecasting methods.

The estimation window runs from 30 April 2021 to 24 July 2021. We use a four-day case-to-hospitalisation lead, a weekly seasonal period, a seven-day forecast horizon, and a 14-day plotting window. Under this specification, a hospitalisation forecast for horizon j depends on

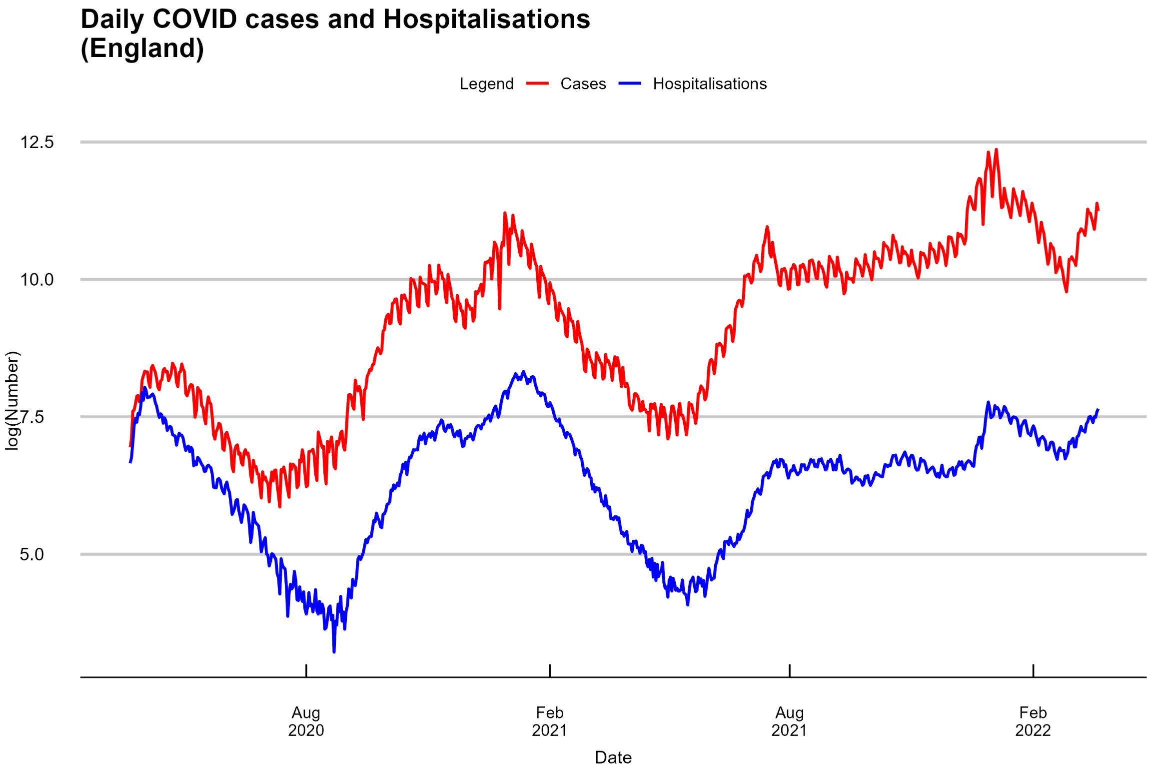

Figure 10: England daily cases and hospitalisations.

the lag-aligned case-growth value at $T + j - 4$. Hence, for $j = 1, \ldots, 4$, the required case-growth values lie within the real-time information set available at the forecast origin T. For $j = 5, \ldots, 7$, the corresponding case-growth values lie beyond T, and are therefore unavailable unless future case values are explicitly supplied.

```
est.start.eng <- as.Date("2021-04-30")
est.end.eng <- as.Date("2021-07-24")
n.lag <- 4
n.forecasts <- 7
confidence.level <- 0.68
plt.length <- 14

mod.eng <- tsgc::SSModelLeadingIndicator(
  Y = eng,
  n.lag = n.lag,
  LeadIndCol = 1,
  sea.period = 7,
  start.date = est.start.eng,
  end.date = est.end.eng
)
res.eng <- estimate(mod.eng)
```

The fitted object `res.eng` can then be used to generate the standard forecast panels.

```
tsgc::plot_log_forecast(
  res.eng,
  Y = eng,
  n.ahead = n.forecasts,
  plt.start.date = est.end.eng - plt.length,
  title = "Forecast of log growth rate of hospitalisations (England)"
)

tsgc::plot_forecast(
  res.eng,
  n.ahead = n.forecasts,
  plt.start.date = est.end.eng - plt.length,
  series.name = "Hospitalisations",
  title = "Forecast of hospitalisations (England)"
)

tsgc::plot_holdout(
  res.eng,
  Y = eng,
  n.ahead = n.forecasts,
  series.name = "Hospitalisations",
  title = "Accuracy: forecast of hospitalisations (England)"
)
```

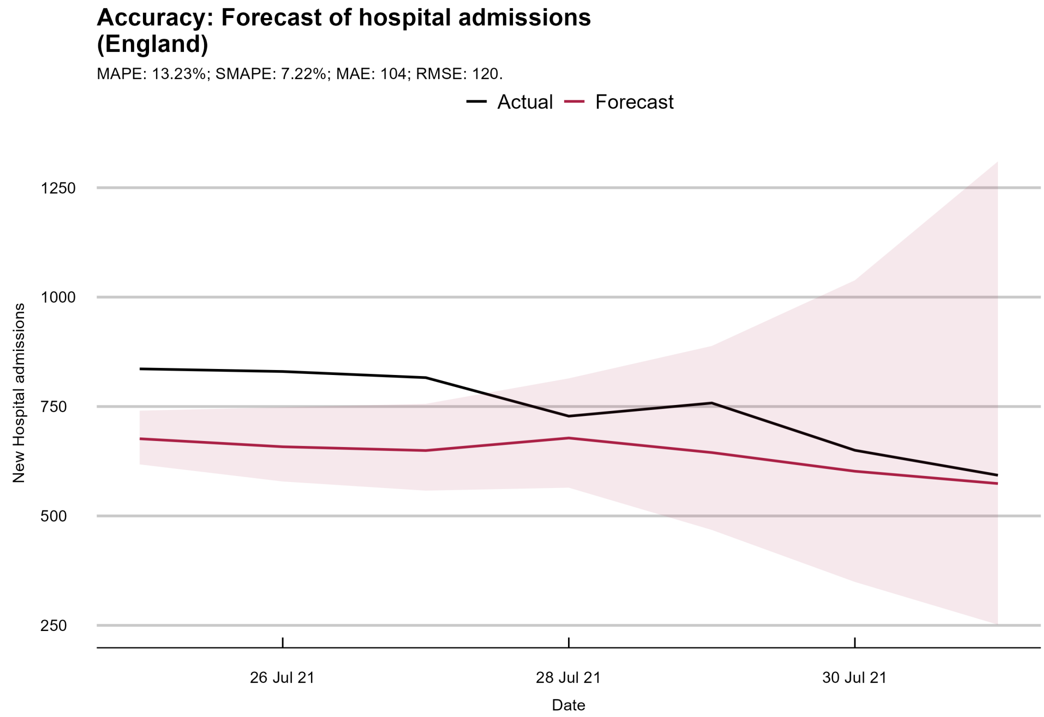

Figure 11: Accuracy of the seven-day forecast of hospitalisations for England.

Tabular forecasts can be extracted directly from the fitted object using `res.eng$predict_level(n.ahead = n.forecasts)`. This baseline England example illustrates the basic leading-indicator workflow: choose a lead series and target series, specify the lag and any seasonal pattern, estimate the model over a fixed sample window, and then obtain level forecasts and holdout diagnostics.

Adding regressors and future covariate paths

The same leading-indicator specification can be enriched with exogenous predictors. In the replication workflow, contemporaneous weather variables—wind speed, relative humidity, temperature, and maximum temperature—are added to both the lead and target equations.

```
data(england_weather_2021, package = "tsgc")
xpred_lead <- xpred_targ <- england_weather_2021[, 1:4]

mod.eng.x <- tsgc::SSModelLeadingIndicator(
  eng,
  n.lag = 4,
  xpred_lead = xpred_lead,
  xpred_targ = xpred_targ,
  sea.period = 7,
  start.date = est.start.eng,
  end.date = est.end.eng
```

```
)
res.eng.x <- estimate(mod.eng.x)
```

When regressors are included, future regressor paths must be supplied before forecasting.

```
supply_xpred.new(res.eng.x, england_weather_2021[, 1:4], idx = "lead")
supply_xpred.new(res.eng.x, england_weather_2021[, 1:4], idx = "targ")
```

The same forecast and holdout panels can then be produced for the augmented model. In practice, the recommended sequence is to establish a sensible lag and seasonal specification first, inspect the resulting forecasts, and only then augment the model with regressors if additional covariates are available and future paths can be supplied.

7.3. UK–Italy: comparing models and selecting lag

The final worked example uses daily COVID-19 cases in Italy and the UK to compare the univariate dynamic Gompertz model with the leading-indicator model and to illustrate lag selection by rolling cross-validation. Italy experienced the first large European outbreak earlier than the UK, making it a natural candidate for a short-horizon lead series during the first wave.

Forecast design and evaluation protocol

To ensure comparability across models and across forecast windows, the design is held fixed throughout this subsection. We use a 14-day forecast horizon, a confidence level of 0.68, and a 30-day plotting window. For the leading-indicator model, weekly seasonality is included via `sea.period = 7`. The baseline case-study lag is $k = 14$ days, while lag selection is performed over the wider grid $k \in \{1, \ldots, 21\}$. Forecast accuracy is assessed using the holdout protocol described earlier and summarised with sMAPE, MAPE, MAE, and RMSE.

Case 1: first-peak window

We first consider the window from 25 February 2020 to 1 April 2020. The UK-only dynamic Gompertz model is estimated on UK daily cases alone.

```
data(ukitaly, package = "tsgc")

n.forecasts <- 14
plt.length  <- 30
CONF        <- CONF_LEVEL

est.start <- as.Date("2020-02-25")
est.end   <- as.Date("2020-04-01")
Yuk       <- ukitaly[, "UK"]

mod_uk_gomp1<- SSModelDynamicGompertz(Y = Yuk, q = q.default,
                                      start.date = est.start, end.date = est.end)
res_uk_gomp1 <- estimate(mod_uk_gomp1)
```

We then estimate the leading-indicator model using Italy as the lead series and the UK as the target series.

```
n.lag <- 14
mod_uk_lead1 <- SSModelLeadingIndicator(Y = ukitaly, n.lag = n.lag,
                                        sea.period = 7,
                                        start.date = est.start,
                                        end.date = est.end)
res_uk_lead1 <- estimate(mod_uk_lead1)
```

The same plotting calls used above can be applied to `res_uk_gomp1` and `res_uk_lead1`. In this first-wave window, the leading-indicator model gives substantially smaller holdout errors than the UK-only dynamic Gompertz model. The holdout summary in Table 2 shows a sharp reduction in sMAPE from 25.58% to 3.74%.

Lag selection via rolling cross-validation

The baseline lag $k = 14$ is motivated by the observed timing of the first-wave evolution in Italy and the UK. To avoid selecting the lag purely by inspection, we use rolling cross-validation over a wider lag grid. The following call evaluates candidate lags over five rolling origins spaced two days apart. Because `SSModelLeadingIndicator` uses one alignment lag per fitted model, the grid search below should be read as model selection over alternative alignments, not as estimation of a distributed-lag response.

```
cv_models <- list()

# Model 1: Vanilla Gompertz
cv_models[["Vanilla_q"]] <- SSModelDynamicGompertz(
  Y = Yuk, q = q.default, start.date = est.start, end.date = est.end)

# Model 2: Vanilla Gompertz with AR(1)
cv_models[["Vanilla_ar1"]] <- SSModelDynamicGompertz(
  Y = Yuk, start.date = est.start, end.date = est.end, ar1 = TRUE)

# Other models: Leading Indicator with different lags from 1 to 21
for (i in 1:21) {
  cv_models[[paste0("Lag", i)]] <- SSModelLeadingIndicator(
    Y = tsgc::ukitaly, start.date = est.start, end.date = est.end, n.lag = i)
}

cross_val(
  Y           = ukitaly,
  model_list  = cv_models,
  est.end.date = est.end,
  criterion    = "smape",
  n.ahead      = 14,
  n.estimate   = 5,
  gap          = 2)
```

For each candidate model, `cross_val()` evaluates forecasting performance over a rolling-origin scheme. The model list includes two UK-only dynamic Gompertz benchmarks and a collection of Italy-to-UK leading-indicator specifications with varying lags.

The first benchmark model, labelled "Dynamic Gompertz, fixed q_ζ", is the target-only model in (2)–(4) with $q_\zeta = 0.005$. The second benchmark model, labelled "Dynamic Gompertz, AR(1) extension", is the dynamic Gompertz model estimated with `ar1 = TRUE`. This benchmark retains the stochastic Gompertz trend and augments it with a latent AR(1) state u_t, which captures short-run serial dependence not absorbed by the local-linear trend. Omitting seasonal terms for notational simplicity, the AR(1)-augmented specification can be written as

$$\log g_t = \delta_t + u_t + \varepsilon_t, \tag{21}$$
$$\delta_t = \delta_{t-1} + \gamma_{t-1}, \tag{22}$$
$$\gamma_t = \gamma_{t-1} + \zeta_t, \tag{23}$$
$$u_t = \rho u_{t-1} + \xi_t, \qquad |\rho| < 1. \tag{24}$$

Thus the two dynamic Gompertz benchmark rows use only the UK target series and have no Italy–UK lag parameter. The remaining models, labelled "Italy lead, $k = \ldots$", are leading-indicator models in which Italian case growth is aligned as a k-day lead for UK case growth.

Table 1 reports an abridged summary of the lag-selection exercise, focusing on the mid-teen lags that perform best. The full $k = 1, \ldots, 21$ grid is reported in the replication material.

Table 1: Abridged cross-validation summary for the UK–Italy comparison (criterion: sMAPE; lower is better). Rows labelled by lag are Italy-to-UK leading-indicator models. The two dynamic Gompertz rows are UK-only benchmarks: the fixed-q_ζ dynamic Gompertz model and the AR(1)-augmented dynamic Gompertz model in (24). The full $k = 1, \ldots, 21$ lag grid is reported in the replication material.

Model class	Specification	Mean sMAPE	Median sMAPE
Leading indicator	Italy lead, $k = 14$	7.43	5.88
Leading indicator	Italy lead, $k = 15$	6.46	**5.26**
Leading indicator	Italy lead, $k = 16$	**6.41**	5.27
Leading indicator	Italy lead, $k = 17$	7.03	6.44
Leading indicator	Italy lead, $k = 18$	7.74	7.70
Leading indicator	Italy lead, $k = 19$	7.48	6.96
Target-only benchmark	Dynamic Gompertz, fixed $q_\zeta = 0.005$	14.22	11.80
Target-only benchmark	Dynamic Gompertz, AR(1) extension	16.52	15.43

The cross-validation results suggest that the best-performing lags in this window lie in the mid-teens. The lowest mean sMAPE occurs at $k = 16$, while $k = 15$ is essentially tied and has the lowest median sMAPE. The baseline $k = 14$ remains competitive. The full lag grid in the replication material gives the same qualitative conclusion: the best lag outside the displayed range is $k = 9$, with mean sMAPE 8.07, which is worse than the mid-teen candidates reported in Table 1. Substantively, the evidence supports a lead of roughly two to two-and-a-half weeks from Italy to the UK over this period. The UK-only dynamic Gompertz benchmarks, including the AR(1)-augmented variant, perform less well than the best leading-indicator specifications in this evaluation window.

Case 2: extended window

We next repeat the same comparison on an extended sample ending on 15 April 2020. The forecasting design is unchanged; only the estimation end date is moved forward.

```
est.start <- as.Date("2020-02-25")
est.end   <- as.Date("2020-04-15")

res_uk_gomp2 <- estimate(
  SSModelDynamicGompertz(
    Y = Yuk, q = q.default,
    start.date = est.start, end.date = est.end)
)

res_uk_lead2 <- estimate(
  SSModelLeadingIndicator(
    Y = tsgc::ukitaly, n.lag = 14,
    start.date = est.start, end.date = est.end)
)
```

The same plotting calls used in Case 1 can then be applied to `res_uk_gomp2` and `res_uk_lead2`. In this extended window, the performance gap narrows. The leading-indicator model remains modestly more accurate than the UK-only dynamic Gompertz model, but both models perform materially worse than in Case 1.

Holdout summary and practical lessons

Table 2 summarizes the holdout metrics for the two models across the two windows.

Table 2: Holdout accuracy summary for the UK–Italy comparison (14-day horizon). sMAPE is emphasised; MAPE, MAE, and RMSE are reported for context.

Window	Model	sMAPE (%)	MAPE (%)	MAE	RMSE
Case 1 (2020-02-25 to 2020-04-01)	Dynamic Gompertz (UK-only)	25.58	79.00	3460	4140
Case 1 (2020-02-25 to 2020-04-01)	Leading indicator (Italy $\rightarrow$ UK, $k = 14$)	3.74	7.50	326	386
Case 2 (2020-02-25 to 2020-04-15)	Dynamic Gompertz (UK-only)	24.97	39.00	1780	1900
Case 2 (2020-02-25 to 2020-04-15)	Leading indicator (Italy $\rightarrow$ UK, $k = 14$)	21.28	34.62	1610	1680

Two messages stand out. First, in the first-peak window the leading-indicator model delivers a substantial improvement over the UK-only dynamic Gompertz model. Second, in the extended window the leading-indicator advantage shrinks, indicating that cross-series lead–lag relationships may be most effective when the leading series is clearly ahead of the target and the linkage remains stable over the forecast horizon.

Taken together, the UK–Italy example suggests a practical workflow: use a leading-indicator model when a credible lead series exists, select the lag empirically over a sufficiently wide grid, and re-check robustness over alternative estimation windows because cross-series relationships can weaken or break over time.

8. Extension to Other Data Frequencies

Although the illustrations in this paper use daily COVID-19 series, the modelling framework implemented in **tsgc** is not tied to a particular observation frequency. The same estimation framework can be applied to regularly sampled cumulative series at weekly, monthly, quarterly, or annual frequency, although the interpretation of the time index, lag, seasonal period, and signal-to-noise ratio changes with the sampling interval. Changing frequency does not alter the underlying estimation procedure; it changes only the interpretation of time. Thus, `n.ahead` is read in periods of the data, any leading-indicator lag is measured in observations at that frequency, and seasonal periods should be specified accordingly.

This feature makes the package applicable beyond epidemic surveillance. Many bounded growth processes, including product diffusion, platform adoption, and sales trajectories, display the same pattern of rapid expansion followed by deceleration towards saturation. Such series can therefore be analysed within the same dynamic Gompertz framework and, where appropriate, with the leading-indicator extension.

The package includes lower-frequency examples that illustrate this point. For instance, `nintendo_sales` contains quarterly console-sales series, while `Plus500` records monthly counts of downloads. In both cases, the workflow is the same as in the daily examples above: select an estimation window, fit the model, choose a forecast horizon, and evaluate the resulting forecasts against held-out observations. No change to the state space specification or estimation algorithm is required.

This ability to work with different observation frequencies broadens the range of applications for **tsgc**. In addition to epidemic monitoring, the package can be used to study product take-up, technology diffusion, and other growth processes for which only lower-frequency data are available.

9. Conclusions

The **tsgc** package is based on a dynamic Gompertz curve model for the log of the growth rate of cumulative cases in an epidemic, with seasonal terms that capture day-of-the-week effects. The estimation is carried out using the **KFAS** package, which implements state-space modelling in R.

The Kalman filter is used to estimate the state vector at each time point. The filter is initialised using a diffuse prior for the initial state vector. We allow the option for the signal-to-noise ratio to either be estimated or fixed at some value based on experience and judgment. Future observations are forecast using predictive recursions.

Epidemics are often characterised by multiple waves. A natural problem in this context is that there is very little data pertinent to the new wave available in its initial stages. The package employs a reinitialisation method using priors in a way that allows data from before the beginning of the new wave to be used in estimation. The package also includes several functions for generating plots and forecasts.

The package is demonstrated using COVID-19 data from South Africa and England, but it can be used to model and forecast any time series variable where a growth curve-like trajectory is expected. Examples might include sales of a new product, innovation adoption, or website traffic.

These strengths should, however, be considered alongside the scope and limitations of the approach. The methods implemented in **tsgc** are intended for short-horizon forecasting of regularly observed cumulative series with positive increments. They are phenomenological time-series models and do not identify causal effects of interventions or represent transmission mechanisms, susceptible depletion, immunity, behavioural change, or policy responses structurally. Forecasts are conditional on the recent stability of the estimated growth dynamics, the chosen signal-to-noise ratio, and, where applicable, the selected reinitialisation date or leading-indicator lag. The incidence bands displayed by the package are state-implied forecast bands rather than fully calibrated predictive intervals, and they do not incorporate uncertainty about model selection, data revisions, future covariate paths, or the generation interval used to compute reproduction-number summaries.

The leading-indicator extension can improve short-horizon forecasts when a stable lead–lag relationship exists, but such relationships may weaken or break under changes in testing, reporting, policy, variants, or behaviour. The empirical examples in this paper demonstrate package functionality rather than providing a comprehensive benchmark across diseases, regions, and epidemic phases.

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Appendix

Incorporating seasonal terms into the state space model

When we add a seasonal term to the model, the observation equation in the dynamic Gompertz curve (2) becomes

$$\ln g_t = \delta_t + \nu_t + \varepsilon_t, \;\; \varepsilon_t \sim NID(0, \sigma_\varepsilon^2), \;\; t = 2, ..., T,$$

where ν_t is the seasonal component, δ_t remains defined by (3), and ε_t remains the iid Normal disturbance.

There are two options for specifying the evolution of the seasonal component. We can either use a trigonometric seasonal or a dummy variable seasonal. In our application, we use a trigonometric seasonal, although the two specifications are closely related. Indeed, Proietti (2000) shows that, under certain conditions, the two approaches are equivalent.

In the trigonometric seasonal approach, the seasonal pattern is represented by stochastic cycles at the seasonal frequencies

$$\lambda_j = \frac{2\pi j}{s},$$

where s is the periodicity of the seasonal effect (Durbin and Koopman 2012). For odd s, the paired trigonometric components are indexed by $j = 1, \ldots, (s-1)/2$. For even s, the paired components are indexed by $j = 1, \ldots, s/2 - 1$, with the final frequency $j = s/2$ corresponding to $\lambda_{s/2} = \pi$. Our daily applications set $s = 7$ to capture day-of-week effects, so the paired frequencies are $2\pi/7$, $4\pi/7$, and $6\pi/7$.

Letting $\nu_{j,t}$ denote the seasonal component associated with frequency λ_j, the seasonal effect is

$$\nu_t = \sum_j \nu_{j,t}, \tag{25}$$

where the sum is over the seasonal frequencies included in the trigonometric representation. For each paired frequency, the state evolves as

$$\nu_{j,t} = \nu_{j,t-1} \cos \lambda_j + \nu^*_{j,t-1} \sin \lambda_j + \omega_{j,t}, \tag{26}$$

$$\nu^*_{j,t} = -\nu_{j,t-1} \sin \lambda_j + \nu^*_{j,t-1} \cos \lambda_j + \omega^*_{j,t}. \tag{27}$$

The disturbances $\omega_{j,t}$ and $\omega^*_{j,t}$ are mutually independent, iid $N(0, \sigma_\omega^2)$ variables. If s is even, the final frequency $\lambda_{s/2} = \pi$ may be represented by the scalar recursion

$$\nu_{s/2,t} = -\nu_{s/2,t-1} + \omega_{s/2,t}.$$

When reinitialising the model with seasonal terms, P_1^r is a block-diagonal matrix based on P_{r+1} which sets the covariances between $(\delta_t, \gamma_t)'$ and $(\nu_{1,t}, \nu_{2,t}, \ldots, \nu_{s^*,t})'$ to be zero. The covariance between δ_t and γ_t, as well as the covariances between $\nu_{1,t}, \nu_{2,t}, \ldots, \nu_{s^*,t}$, are permitted to be non-zero and come directly from P_{r+1}.

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